

MacroABM-CA: An Open-Source Regional Agent-Based Model of Canada, with an Application to Investment Tax Credits

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Abstract

We introduce MacroABM-CA, to our knowledge the first open-source regional macroeconomic agent-based model (ABM) of Canada. The model represents the Canadian economy as thirteen provinces and territories and fifty industries, and it is initialized from a newly constructed 2022 interprovincial input-output system and Canadian household microdata. Because firms and households make plans that may or may not be realized in decentralized markets, the model can show where a national policy enters the economy, which provinces and industries absorb it, and how its effects persist. In a historical comparison over 2022-2024, the model approaches a simple first-order autoregressive (AR(1)) benchmark for quarterly output growth and improves on it for unemployment. We demonstrate the framework with a stylized 30 percent investment tax credit (ITC) inspired by Canada's Clean Technology ITC. In constant prices, each dollar of gross refunds during 2026-2030 is associated with about \$0.73 of additional investment and \$0.98 of additional gross value added (GVA). Because the eligible machinery is largely imported, imports rise by about \$0.32 per dollar of refunds; most of the response is concentrated in Ontario, Alberta and Quebec, and the effect persists after the credit expires because firms adapt their demand expectations to the activity the policy has generated. These results matter for Canadian policy because debate in a federation turns on where effects arise and to whom, not only on their aggregate size; MacroABM-CA provides an open and reproducible platform for tracing that regional transmission.

Keywords: agent-based macroeconomics; regional modelling; input-output; investment tax credit; adaptive expectations; energy and industrial policy.

JEL: C63, E22, H25, R13, Q48.

1 Introduction

Governments increasingly use targeted fiscal and industrial instruments, such as investment tax credits for clean technologies, to shape where and how firms invest. The effects of such policies depend on which firms respond, what they buy, where their suppliers are located and how the resulting income is spent. These features are difficult to represent in models built around a single representative firm and household. This paper introduces MacroABM-CA, an open-source regional

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macroeconomic agent-based model of Canada designed to trace how a national policy is transmitted through heterogeneous firms, households and provincial economies, and demonstrates it with a stylized investment tax credit.

1.1 Agent-based models for macroeconomic and policy analysis

Quantitative policy analysis has long relied on computable general equilibrium (CGE) and dynamic stochastic general equilibrium (DSGE) models. These models are internally consistent and well suited to comparing equilibrium outcomes, but they typically assume optimizing representative agents, market clearing and a unique equilibrium path. They are therefore less suited to questions in which heterogeneity, rationing, bounded rationality and out-of-equilibrium adjustment determine the outcome (Farmer and Foley 2009; Stiglitz 2018). Agent-based models take a different approach. They simulate many interacting agents that follow explicit behavioural rules and transact in decentralized markets, so that aggregate outcomes emerge from individual decisions rather than being imposed by an equilibrium condition (Dawid and Delli Gatti 2018; Haldane and Turrell 2018; Axtell and Farmer 2025). Their main strengths are the ability to represent heterogeneous agents and networks, to distinguish what agents plan from what markets allow them to realize, and to show how local decisions add up to aggregate dynamics. Their main weaknesses have been a large number of behavioural parameters, a historically weak link to data and correspondingly difficult validation (Fagiolo, Moneta, and Windrum 2007).

A recent generation of data-driven macroeconomic ABMs addresses these weaknesses directly by initializing agents from national accounts, input-output tables, administrative statistics and household microdata, and by evaluating models against observed macroeconomic series. Central banks have been prominent in this development: the Bank of England has used ABMs to study housing markets and macroprudential policy (Baptista et al. 2016; Turrell 2016), and the Bank of Canada has developed CANVAS, a data-driven ABM for Canadian forecasting and monetary-policy analysis (Hommes et al. 2025).

ABMs have also become established in energy and climate-policy research. Reviews by Hesselink and Chappin (2019) and Castro et al. (2020) emphasize their ability to incorporate bounded rationality, heterogeneous preferences, peer effects and institutional constraints that a single optimizing agent cannot capture, and Rai and Henry (2016) show how social interaction shapes technology adoption. At the macroeconomic scale, agent-based integrated assessment and stock-flow models have been used to study the interaction of climate damages, green fiscal policy and green finance with growth and employment (Balint et al. 2017; Lamperti et al. 2018; Monasterolo and Raberto 2018), and ABMs have been proposed as a way to represent the structural and distributional dimensions of the low-carbon transition that equilibrium models tend to abstract from (Farmer et al. 2015). Related ABM work shows that a uniform initial policy exposure need not imply a uniform outcome when agents occupy different structural positions: Rio-Chanona et al. (2021), for example, find that occupational mobility networks produce very different employment trajectories for workers facing similar automation exposure.

1.2 The data-driven MacroABM lineage

MacroABM-CA belongs to this data-driven lineage. Poledna et al. (2023) constructed an agent-based model of Austria initialized from national accounts, input-output tables and microdata and showed that it can forecast competitively with standard time-series and DSGE benchmarks. The MacroABM framework developed at INET Oxford by Wiese et al. (2024) extends this architec-

ture with realistic synthetic populations, richer household and firm behaviour, housing and credit markets and a modular, open implementation; applied to 38 OECD economies, it produces out-of-sample forecasts that compare favourably with both the earlier model and standard benchmarks. CANVAS (Hommes et al. 2025) independently adapts the Poledna architecture to Canada for national forecasting and monetary-policy analysis. MacroABM-CA builds directly on the software architecture of Wiese et al. (2024) but pursues a different objective: representing how national fiscal and industrial policies are transmitted across provinces, territories and industries.

1.3 Why a regional model of Canada

A nationally uniform policy does not meet a uniform Canadian economy. Provinces differ substantially in industrial composition, labour-market conditions, capital intensity, import dependence and reliance on suppliers in other provinces, and Canadian internal-trade research documents significant provincial border frictions and strong interdependence among industries (Albrecht and Tombe 2016; Bemrose, Brown, and Tweedle 2020; Lapham and Teeter 2023). Production-network research shows more generally that the aggregate effect of a shock depends on the position of the affected industries and on their input and market linkages (Acemoglu et al. 2012; Pichler et al. 2022). An investment incentive received by a firm in one province may therefore generate demand for imported machinery, intermediate production in another province, employment income locally and subsequent household spending across several sectors. A national model can capture the aggregate outcome without revealing where the policy enters, where it meets constraints or how its effects are distributed geographically. Provinces also set many of their own policies, and a regional model allows province-specific instruments or parameters to be represented, although the experiment in this paper applies a single federal credit uniformly.

Multi-region input-output and CGE analyses have combined Canadian regional accounts with international trade data (Bachmann, Roorda, and Kennedy 2015), but data limitations commonly force coarse sectoral aggregation or proportional assumptions when allocating imports and interregional trade, which can obscure the true geographic and sectoral origins of intermediate inputs (Schulte, Jakobs, and Pauliuk 2021).

1.4 Investment incentives as a demonstration

Investment tax credits illustrate why this combination of heterogeneity, market realization and regional structure matters. The classical rationale is that a credit lowers the user cost of capital and so raises firms' desired investment (Hall and Jorgenson 1967). Empirical work, however, finds substantial heterogeneity in how firms and sectors respond to investment incentives, reflecting differences in eligibility, financial position, project timing and adjustment costs (Zwick and Mahon 2017; Ohrn 2019; Maffini, Xing, and Devereux 2019), and recent evidence shows that effects extend beyond the subsidized firm through its input-output links (Lerche 2025). An aggregate investment equation can summarize the average response but cannot show which firms respond, whether their orders are filled, where the equipment is produced or how the resulting income circulates. An ABM with an explicit regional production network can.

In MacroABM-CA the behavioural margin is stated openly rather than estimated internally: the credit reduces the effective user cost of a firm's investment in proportion to the eligible share of its planned capital, and an externally specified elasticity converts that reduction into additional desired investment. The model then follows the impulse as desired investment is purchased in goods markets, installed as productive capital and incorporated into later production decisions,

while realized sales update firms’ demand expectations. The policy is transmitted through the sequence

plans → market realization and feasibility → production and sales → state and expectation updates → subsequent plans,

so a larger desired capital order is not necessarily a realized purchase, a realized purchase does not immediately raise output, and higher output can affect later activity through employment, household income, sales and expectations.

1.5 Contributions and main findings

Against this background, MacroABM-CA brings together four elements that are usually studied separately: a data-driven macroeconomic ABM, a detailed Canadian synthetic household population, a new province-by-sector interprovincial input-output system and a firm-level investment-policy experiment. The paper makes three contributions.

1. **An open regional model and dataset.** MacroABM-CA regionalizes the Wiese et al. (2024) framework across thirteen provinces and territories and fifty industries, producing 650 representative province-sector firms. It is initialized from a newly constructed 2022 interprovincial input-output system, built from Statistics Canada’s provincial supply-use tables and bilateral interprovincial trade and harmonized with the fifty-sector OECD Inter-Country Input-Output classification, which allows the Canadian regional system to be linked to international trade flows and compared directly with other economies. Bilateral province-to-province trade by commodity is observed directly; only its split across using industries is allocated, under an origin-homogeneity assumption described in Section 2.1 and Appendix B.3. To our knowledge it is among the first such systems assembled at this resolution for public use. The model code, data-construction procedures, configurations and experiment code are available at <https://github.com/uvic-sesit/macroabm-ca>.
2. **A framework that makes transmission observable.** Because planned and realized quantities are distinguished throughout, the model identifies where a policy impulse is absorbed or amplified, why responses differ across province-sector firms and which effects persist after an intervention ends.
3. **A transparent policy application.** We trace a stylized ITC from its initial behavioural effect through market realization, regional propagation and subsequent adjustment, and separate the contemporaneous investment response from the part that arises because the policy has already changed demand, production, capital stocks and expectations. This distinction matters for policy evaluation because it shows whether effects come from the incentive itself or from the economy’s subsequent response to it, and therefore how long they last once the incentive is withdrawn.

The application applies a 30 percent refundable credit to a broad machinery-and-equipment proxy from 2026 through 2030. It is a general demonstration rather than a statutory evaluation of the Canadian programme. Four findings stand out.

1. **Scale.** In constant prices, each dollar of gross refunds is associated with about \$0.73 of additional investment, \$2.07 of additional gross production and \$0.98 of additional GVA during the policy window. Investment rises about 1.6 percent above the no-policy path. Production exceeds GVA because it also counts the intermediate transactions generated as activity moves through supply chains.

2. **Leakage.** Both eligible machinery sectors are net importers, so a substantial part of the stimulus flows abroad: imports rise by about \$0.32 per dollar of refunds.
3. **Persistence.** More than half of the production and GVA response during the policy window, and more than four-fifths after it, comes from firms adapting their demand expectations to the higher sales the policy generated. A temporary credit therefore keeps affecting activity after it expires.
4. **Regional concentration and heterogeneity.** Ontario, Alberta and Quebec receive about three-quarters of refunds and generate about three-quarters of the additional GVA, largely because of their size, while some smaller provinces respond more strongly relative to their economies.

These are treatment-control differences within the model economy, not forecasts of the actual programme, and they are conditional on the stated behavioural assumptions (Section 7).

1.6 Structure of the paper

Section 2 presents the model architecture and simulation sequence. Section 3 describes the validation strategy and historical performance. Section 4 introduces the Canadian policy context and the design of the experiment. Section 5 reports the aggregate, dynamic, regional, monetary and fiscal results. Section 6 draws out policy implications, Section 7 discusses limitations and Section 8 concludes. The appendices document implementation, data construction, validation detail, the policy mechanism, regional heterogeneity and robustness.

2 Model Architecture and Simulation

This section describes the model as a general platform; the investment tax credit is introduced separately in Section 4. MacroABM-CA preserves the parent framework’s central distinction between planned and realized activity. Agents make decisions using inherited states and adaptive expectations, while employment, credit, purchases and sales emerge through decentralized markets and may fall short of plans. The economy is therefore generated recursively rather than solved as a simultaneous equilibrium. This section describes the mechanisms needed to interpret the policy experiment; detailed algorithms, balance sheets and parameter provenance are reserved for the appendices.

2.1 Regional structure and model resolution

The Canadian economy is represented by thirteen provinces and territories and fifty industries in a common OECD ICIO classification. The policy experiments use one producing firm in each province-sector cell, yielding 650 representative producers. The broader MacroABM architecture can accommodate several heterogeneous firms within a cell, that is, multiple firms of different size, productivity or technology operating in the same industry and province. That resolution is not used here because disciplining a within-cell firm distribution requires confidential Canadian firm-level microdata, whose integration is under way as separate work. In the reported experiments, heterogeneity across firms is therefore structural: it reflects differences in regional scale, sectoral technology, labour use, capital composition and sourcing rather than a simulated distribution of establishments within each industry and province.

Production and trade relationships are initialized from a newly constructed 2022 Canadian inter-provincial input-output system. Statistics Canada’s provincial supply-use and interprovincial trade

data are reconciled to the fifty-sector OECD classification and connected to the rest of the world. Bilateral province-to-province trade in each commodity is observed directly. What is not observed is how a province’s imports of a commodity from each partner province are split across the industries that use them; we allocate this split under an origin-homogeneity assumption, namely that all using industries in a destination province draw on the same mix of origin provinces for a given commodity (Appendix B.3). This system determines the technical coefficients for intermediate inputs and for capital inputs, and the initial sourcing shares within provinces, across provinces and internationally. In the present experiments, supplier topology and technical coefficients are fixed: firms transact over existing links and do not endogenously adopt a new technology or create suppliers. Quantities ordered and realized over those links remain endogenous. The same policy can consequently generate different regional outcomes because firms differ in scale, input composition, available capacity and domestic or foreign sourcing. Appendix B documents the 2022 provincial and sectoral economic structure that this initialization encodes (Table A2 and Figure A1 for provinces, Figure A2 for sectors), including the provincial distribution of value added and final demand, provincial trade balances, and the sectoral trade exposure of the ITC-eligible machinery sectors.

2.2 Agents, institutions and balance-sheet relationships

Table 1 summarizes the active agents and institutions. Firms produce, employ workers, purchase inputs, set prices and maintain physical and financial positions. Individuals participate in the labour market and are grouped into households that receive income, hold assets and liabilities, and plan consumption. Commercial banks accept deposits, extend several loan types and adjust rates in response to the common policy rate. A synthetic government in each region buys goods, collects represented taxes, pays cash benefits and ITC refunds, and records deficits and debt. Monetary-policy objects are synchronized across regions to represent a common national setting. The rest of the world supplies imports and demands Canadian output. These balance-sheet and transaction structures are inherited principally from Wiese et al. (2024), with the Canadian modelling choices described below. Each individual agent represents 1,000 persons in the ten provinces and 10 persons in the three territories, so that the territories retain enough agents for functioning labour and goods markets; all monetary quantities and national head counts are scaled accordingly.

Agent or institution	Number in the model	Endogenous decisions and transactions each quarter	State variables carried between quarters
Province-sector firms	650 (13 regions $\times$ 50 industries)	Production targets; hiring; intermediate and capital orders; prices; credit applications	Demand history, inventories, input and capital stocks, deposits, loans and equity
Individuals and households	About 35,000 labour-force agents in 2022, grouped into households representing 15.455 million Canadian households	Labour supply and job acceptance; consumption and housing-related plans and realized purchases	Employment, wages, benefits, saving propensities, deposits, debt, housing and financial assets
Commercial banks	One representative bank per region (13)	Loan supply and allocation; lending, deposit and overdraft rates	Deposits, loan books, equity, liquidity and non-performing loans

Agent or institution	Number in the model	Endogenous decisions and transactions each quarter	State variables carried between quarters
Synthetic regional governments	13	Purchases; represented taxes; cash transfers; policy payments such as tax-credit refunds; deficit and debt accounting	Tax parameters, benefit levels and government debt
National monetary block	1	Common policy-rate environment	Policy rate and inflation target
Rest of world	1	Import supply and demand for Canadian exports	Foreign demand, prices and bilateral trade shares

Table 1. Agents and institutions in MacroABM-CA: their number, the decisions and transactions they make endogenously in each quarter, and the state variables they carry from one quarter to the next. Production targets are listed as decisions because they are chosen by the firm; realized production is an outcome of those decisions and of market constraints, and is recorded in the firm’s state.

The model records linked financial positions for households, firms and banks. Its fiscal perimeter is less complete: government debt is an accounting stock rather than a security issued through an explicit bond market and held by another agent. Regional governments should therefore be interpreted as synthetic consolidated fiscal units, not literal provincial treasuries. Absolute deficit and debt paths are used diagnostically rather than as Canadian fiscal forecasts.

2.3 Information, expectations and quarterly timing

One model period is one quarter, and each quarter passes through four stages that together form a cycle (Figure 1). In **Stage 1 (planning)**, agents start from the balance sheets, employment relationships, inventories, input stocks, prices and realized demand inherited from the previous quarter, update their expectations and formulate production and input plans. In **Stage 2 (labour matching and production)**, labour matching determines employment, after which firms produce using previously installed capital and intermediate stocks and set prices. In **Stage 3 (market realization)**, firms, households and governments place orders, eligible agents apply for credit, and credit and goods markets determine realized purchases, sales and imports. In **Stage 4 (accounting and state update)**, incomes, taxes, transfers, financial positions, inventories, capital stocks and demand histories are updated and carried into the next quarter’s planning. This recursive sequence follows the planning-market-accounting structure of Wiese et al. (2024) and Poledna et al. (2023); Appendix A.1 lists the full event sequence. Sections 2.4 to 2.11 describe the mechanisms at each stage, beginning with firms.

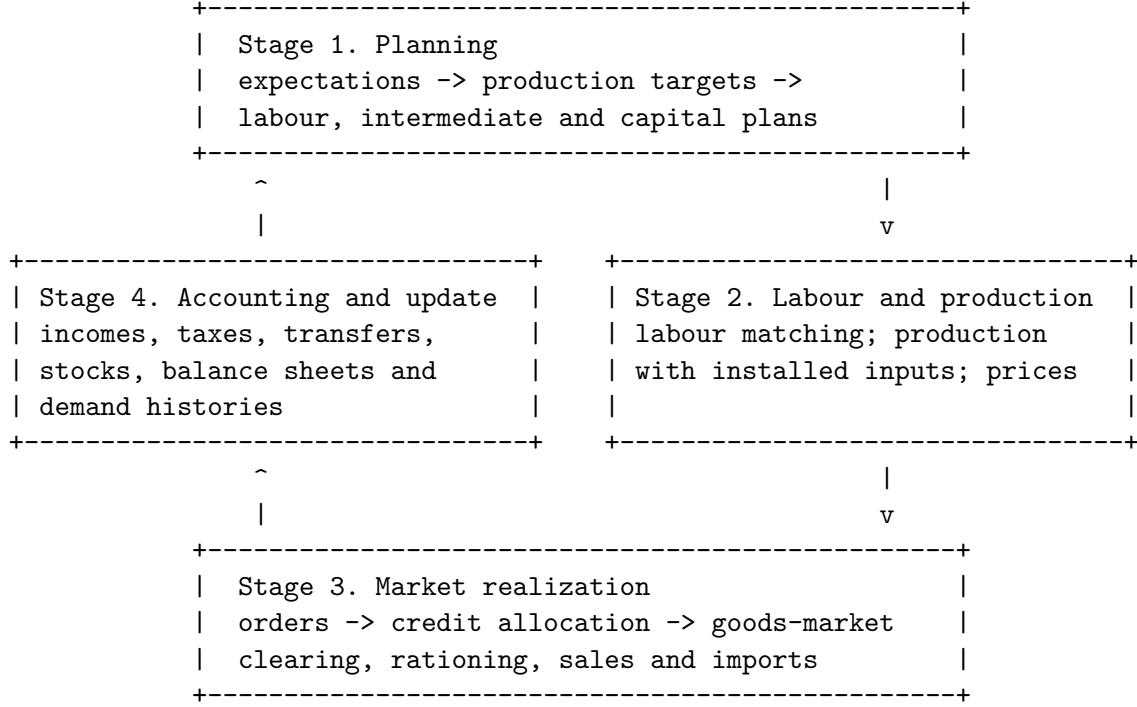


Figure 1. The quarterly cycle of MacroABM-CA. Each quarter moves from planning (Stage 1) through labour matching and production (Stage 2) and market realization (Stage 3) to accounting and state updating (Stage 4), whose results become the starting point for the next quarter's plans. The diagram applies to all agents; firm-specific steps are described in Sections 2.4 to 2.6. The full ordered event list appears in Appendix A.1.

This ordering has an important implication: capital purchased in Stage 3 of quarter t is installed in Stage 4, after that quarter's production, so it raises feasible output only from the following quarter onward. Expectations use observed histories and model-generated macroeconomic signals rather than perfect foresight.

2.4 Firm expectations and production planning

Firm i 's demand expectation combines a smoothed history of observed demand with aggregate and firm-specific growth signals, adapting the parent model's forecasting rule to the Canadian application (Wiese et al. 2024):

$$\bar{D}_{i,t} = (1 - \alpha)\bar{D}_{i,t-1} + \alpha D_{i,t-1}, \quad \hat{D}_{i,t} = (1 + \phi_g \hat{g}_t)(1 + \phi_i \hat{g}_{i,t})\bar{D}_{i,t}, \quad (1)$$

Here $D_{i,t-1}$ is the demand recorded once the preceding quarter's goods market has cleared, and $\bar{D}_{i,t}$ is its smoothed history. The parameter α is the demand-smoothing (adaptation) speed: the weight a firm places on the most recent quarter's demand relative to its accumulated history, so a larger α means expectations respond faster to new information. The terms $\hat{g}_t$ and $\hat{g}_{i,t}$ are aggregate and firm-specific growth signals, respectively, and ϕ_g and ϕ_i are the weights on those two signals. The firm-specific signal is a single composite index that combines two pieces of information, the firm's balance of demand against supply and its price relative to its sector, so that a firm facing excess demand or charging below its sector's price expects faster growth. A fraction of unmet orders

also enters recorded demand; the weight on unmet orders is listed in Table A1 (Appendix A.2). Expectations are therefore adaptive: they are formed from past observations and neither anticipate future policy nor solve a forward-looking optimization problem.

The reported specification sets $\alpha = 0.40$. This value is a historically assessed configuration choice rather than a formally estimated Canadian parameter. It balances responsiveness to recent realized demand with smoothing of short-run fluctuations.

Expected demand is translated into target production. Although the inherited architecture permits inventory and financial corrections, the baseline used here sets target production equal to expected demand:

$$Q_{i,t}^* = \max\{\varepsilon, \widehat{D}_{i,t}\}, \quad (2)$$

Inventories still contribute to goods available for sale and are updated after transactions, but do not directly alter the target. Target production is thus a desired operating scale, distinct from physical production and realized sales.

2.5 Input demand, investment and production feasibility

Firms produce with a Leontief (fixed-proportions) technology: output requires labour together with intermediate inputs and capital inputs in fixed ratios, so inputs cannot substitute for one another. The intermediate-input coefficients and capital-input coefficients are obtained from the input-output system. In the reported specification, desired labour and intermediate inputs follow target production without being reduced in advance to the output supported by inherited capital. This preserves the distinction between the production plan and its subsequent feasibility: inherited capital constrains actual production, not the firm's initial labour and intermediate plans.

Desired gross capital purchases combine replacement with partial adjustment toward a reference stock. Let $\widetilde{Q}_{i,t}^K = \min\{Q_{i,t}^*, \mathcal{C}_{i,t-1}^K\}$, where $\mathcal{C}_{i,t-1}^K$ is output feasible with inherited capital, and let $K_{ij,t}^{\text{ref}} = K_{ij,t-1}(Q_{i,t}^*/Q_{i,t-1})$. Baseline desired purchases are

$$I_{ij,t}^{d,0} = \max\{0, d_{ij}\widetilde{Q}_{i,t}^K + \lambda_K(K_{ij,t}^{\text{ref}} - K_{ij,t-1})\}, \quad (3)$$

where d_{ij} is the replacement requirement and λ_K governs stock adjustment. This is a demand-led accelerator rather than an intertemporal optimizing condition. When a firm's previous output is zero the growth ratio in K^{ref} is set to one, so the reference equals the inherited stock and the adjustment term vanishes; when the inherited stock of a required capital type is itself zero, the reference reverts to a base-year rule so that the stock can be rebuilt (Appendix A.3). This equation describes the model without any policy. The investment tax credit introduced in Section 4.3 enters by scaling the requirement term before stock adjustment; it does not directly alter Q^* , productivity or technical coefficients.

Actual production is constrained by employment after labour matching and by input stocks installed before current production. Let Q^L , Q^M and Q^K denote labour-, intermediate- and capital-feasible output. Under the Leontief production technology inherited from Poledna et al. (2023) and Wiese et al. (2024),

$$Q_{i,t} = \min \{Q_{i,t}^*, Q_{i,t}^L, Q_{i,t}^M, Q_{i,t}^K\}, \quad Q_{i,t}^M = \min_j \{a_{ij}^M M_{ij,t-1}\}, \quad Q_{i,t}^K = \min_j \{a_{ij}^K K_{ij,t-1}\}. \quad (4)$$

The coefficients a_{ij}^M and a_{ij}^K are output-per-unit-input coefficients: feasible output is the binding minimum $\min_j \{a_{ij} \text{stock}_{ij}\}$, so input use is $Q_{i,t}/a_{ij}$, and they are the reciprocals of conventional input-per-output requirements. The requirement d_{ij} in Eq. (3), by contrast, is an input-per-output replacement and depreciation coefficient. Capital and intermediate types the firm does not require carry an infinite productivity coefficient and are excluded from the binding minimum. Intermediate goods are used in fixed proportion to output and capital depreciates according to the capital-requirement matrix. The smallest term in (4) identifies the contemporaneous production constraint, not the full contribution of accumulated capital. Investment can affect future capacity, complementary input demand and subsequent plans even when capital is not the binding constraint in a given quarter.

2.6 Markets, matching and rationing

Desired quantities need not be realized. This distinction is central to the Wiese-Poledna architecture and to the aggregation of local decisions.

Before quarterly matching, labour-force membership is updated using annual labour-force levels aligned to the corresponding calendar year. Firms then compare desired with current employment and create vacancies or release workers. Matching permits mobility across industries and accounts for reservation wages and worker availability, so employment plans may remain unrealized.

The goods market clears intermediate, capital and final-demand orders against production, inventories and foreign supply while respecting observed sourcing patterns. When supply is insufficient, orders are rationed. Realized purchases, rather than desired orders, enter input stocks, household and government expenditure, and any policy payments tied to purchases, such as the tax-credit refunds of Section 4.

Credit is allocated subject to bank capital, portfolio limits, borrower conditions and loan supply. Firms seek short-term finance for intermediate inputs and longer-term finance for investment, while households may seek consumption or mortgage credit. Credit and interest payments affect balance sheets and profits, but credit rationing does not directly scale firms' current input orders. Its immediate quantitative restriction on investment is therefore limited.

Stochastic elements in price adjustment and labour matching are held common across matched baseline and policy simulations. Appendix A.1 places these market steps in the full quarterly event sequence.

2.7 Labour, households and consumption feedback

Individuals supply labour according to their activity status and receive wages when employed or unemployment benefits when eligible. Employment relationships formed in the labour market determine firm labour input and household employee income. Households also receive other cash transfers and financial or rental income. Their saving propensities and sectoral expenditure weights are heterogeneous and inherited from the synthetic-population initialization.

For the policy experiments, household consumption responds endogenously to disposable income, strengthening the feedback from employment and transfers to final demand. This behaviour follows the household architecture of Wiese et al. (2024). In compact form,

$$Y_{h,t}^d = \max\{0, Y_{h,t}^e - \tau_y[(1 - \tau_{si})W_{h,t} + Y_{h,t}^f] - \tau_{si}W_{h,t}\}, \quad C_{h,t}^d = \frac{(1 - s_h) \max\{B_{h,t}, Y_{h,t}^d\}}{1 + \tau_{vat}}, \quad (5)$$

where expected household income Y^e comprises expected employee income, expected social transfers, rental income and expected financial-asset income; W is employee income, Y^f taxable financial income, $B_{h,t}$ benefit income and s_h the household saving rate. Desired consumption is the fraction $(1 - s_h)$ of the greater of disposable income and the benefit floor, deflated by the consumption tax and allocated across sectors using household-specific expenditure weights. In the reported specification the consumption-smoothing weight is set to zero, so consumption tracks current disposable income rather than a moving average; realized purchases may still be rationed in the goods market. Employment and wages can therefore affect taxes, transfers, disposable income, final demand and subsequent firm expectations. Household investment follows an exogenous path. The general three-branch consumption rule and its parameters are documented in Appendix A.3.

2.8 Government and fiscal dynamics

Government target purchases are exogenous and allocated across industries using the initialized government final-demand composition. Government is nevertheless a participant in the goods market: realized purchases can differ from the target because available supply is shared among buyers. Writing $G_{r,t}^d$ for region r 's target (desired) purchases in quarter t and $G_{r,t}^r$ for its realized purchases, $G_{r,t}^r \neq G_{r,t}^d$ in general, and only realized purchases enter expenditure accounting.

The fiscal block collects the taxes represented in the model: taxes on production, consumption, capital formation and exports; corporate taxes on firms and banks; personal income tax; employee and employer social contributions; and public-housing rental income. It pays unemployment benefits and other public cash social benefits. For the Canadian application, aggregate transfers are initialized using OECD SOCX public *cash* expenditure rather than total public social expenditure, excluding in-kind services that would overlap conceptually with government consumption. Existing allocation rules distribute unemployment payments to unemployed individuals and other transfers across households.

When a refundable investment tax credit is active (Section 4), the government refunds the statutory share of firms' realized eligible investment expenditure; the refund, $ITC_{r,t}$, is credited to recipient firms and included once in government expenditure, and it is zero in the absence of the policy. Let $B_{r,t}$ denote cash benefits paid, $R_{r,t}$ represented tax revenue, $L_{r,t}$ government debt and r_t^q the quarterly interest rate on that debt. Regional deficit $DEF_{r,t}$ and debt then follow

$$DEF_{r,t} = G_{r,t}^r + B_{r,t} + ITC_{r,t} + r_t^q L_{r,t-1} - R_{r,t}, \quad L_{r,t} = L_{r,t-1} + DEF_{r,t}, \quad (6)$$

No residual revenue item or fiscal reaction rule is imposed, so the investment tax credit is financed entirely by a higher deficit and debt rather than by an offsetting tax increase or spending cut. Because the expenditure, revenue and debt series do not share a fully harmonized Canadian institutional perimeter, and government debt has no explicit bondholder, absolute deficit and debt levels are diagnostic. Gross ITC refunds and matched treatment-control changes in represented taxes, benefits, purchases and interest are the relevant fiscal outputs.

2.9 Banks, credit and monetary policy

Commercial banks inherit deposits and loan portfolios and supply firm loans, household consumption loans and mortgages. Loan allocation is restricted by bank equity, capital adequacy, portfolio conditions and borrower balance sheets. New lending rates respond to the common policy rate through the pass-through and error-correction structure developed by Wiese et al. (2024); deposit and overdraft rates also respond to the common rate. Existing loans accrue interest at their contracted model-period rates.

Because the model is quarterly, externally quoted annual interest rates are converted to quarterly effective rates before they are applied to financial stocks; the conversion and contract conventions are documented in Appendix A.3.

The model accommodates alternative monetary environments. The main environment holds the policy rate constant, while an alternative endogenizes the common rate through a smoothed, inflation-only Taylor-type rule with interest-rate smoothing, an inflation response, a neutral real rate and an inflation target; its exact form and parameters are given in Appendix A.3. Poledna et al. (2023) provide the precedent for Taylor-type monetary policy in this model lineage, while Wiese et al. (2024) develop the MacroABM implementation and estimated bank-rate pass-through. Changes in the common rate are transmitted to bank lending, deposit and overdraft rates and can affect credit allocation, interest cash flows and aggregate demand. The alternative Canadian rule is a transparent stylization rather than Wiese et al.’s estimated policy equation or a calibrated Bank of Canada reaction function. Because firms’ desired-capital equation contains no direct policy-rate term, monetary transmission to investment operates indirectly through bank rates, credit outcomes, cash flows and demand.

2.10 Prices and inflation

Firms adjust prices in response to expected producer-price inflation, firm-level demand pressure, production costs and an idiosyncratic innovation, following Wiese et al. (2024):

$$p_t = p_{t-1}(1 + \epsilon_t^p)(1 + \gamma_g \hat{\pi}_t^{PPI}) \times (1 + \gamma_d z_t^d)(1 + \gamma_c z_t^c), \quad (7)$$

Here z^d measures excess demand, conditional on the firm’s price relative to its sector, while z^c compares unit costs with sectoral prices. The baseline retains expected inflation and idiosyncratic innovations, permits demand-pull adjustment and excludes direct cost-push adjustment. The demand-response coefficient is a Canadian modelling choice rather than a structural estimate; its value is reported in Table A1 (Appendix A.2).

CPI, PPI and the capital-formation price index aggregate sectoral prices using expenditure or production weights. Their levels reflect aggregate inflation expectations, firm-level innovations and demand pressure. Treatment-control inflation differences therefore provide the more informative measure of ITC-induced price pressure.

2.11 Interprovincial and international transmission

Every buyer’s sectoral demand is routed through the 2022 sourcing structure. A firm can purchase from its own province, another province or the rest of the world; households and governments similarly generate sector-specific final demand. Canadian suppliers can sell locally, interprovincially

or to foreign demand. Realized imports and interprovincial deliveries depend on the interaction of these sourcing shares with current orders, prices and available supply.

External demand and labour-force paths remain exogenous in the policy experiments. Province-to-rest-of-world demand follows the projection assumptions while the underlying interprovincial topology remains unchanged. The model can therefore measure the foreign-sourcing response and domestic or interprovincial propagation, but not endogenous trade-link formation or a structural change in sourcing shares caused by the ITC. Regional spillovers arise from quantities transacted over the fixed network and from subsequent updates to incomes, demand and firm plans.

2.12 Initialization, parameterization and calibration

Parameter provenance is organized in three layers. Where mechanisms are unchanged, MacroABM-CA retains the calibrated or empirically specified parameterization of Wiese et al. (2024). That study formally calibrates a limited subset of parameters using simulation-based Bayesian methods and fixes others from empirical evidence or modelling assumptions. Canadian data are then used to initialize the regional economy, while parameters introduced or modified for the Canadian and ITC applications are disciplined using Canadian historical diagnostics and the empirical policy literature. MacroABM-CA thus combines inherited behavioural parameterization, Canadian empirical initialization and application-specific assumptions; it does not imply that every parameter was newly estimated for Canada.

Parameter layer	Content	Status in MacroABM-CA
Inherited model parameters	Production, matching, market clearing, balance sheets, bank-rate pass-through and adaptive rules	Retained from Wiese et al. (2024) and, through it, Poledna et al. (2023), distinguishing formally calibrated parameters from fixed or literature-based settings
Canadian initialization	Province-sector accounts, technical coefficients, sourcing, population, household finances, labour and fiscal initial conditions	Constructed from Canadian data and mapped to the model’s common classifications
Canadian and policy-specific parameters	Household and fiscal specification, adaptive speeds, capital and price adjustment, ITC eligibility, user-cost elasticity and policy timing	Informed by Canadian diagnostics and the empirical policy literature; scenario assumptions are assessed through sensitivity analysis

The synthetic population preserves heterogeneity in demographic, labour-income, expenditure and financial characteristics, with household weights linking simulated agents to regional population and labour-force totals. Table A1 in Appendix A.2 lists the parameters of the specification used in this paper and classifies each as data-implied, inherited from a formal calibration in the parent model, inherited as a fixed model assumption, selected through Canadian validation, or imposed for the policy experiment; Appendix B.3 (Table A3) documents how the input-output data map into the model and the allocation assumptions involved. The key distinction is that production coefficients and bilateral sourcing shares are taken directly from data, whereas adjustment speeds and response elasticities are behavioural parameters that are inherited or chosen.

In terms of the quarterly cycle of Figure 1, any change to firms’ plans made in Stage 1 is filtered through labour matching and production in Stage 2 and through finance, suppliers and imports in Stage 3; the resulting capital, income and sales are recorded in Stage 4 and shape the next quarter’s plans. A temporary shock to plans can therefore affect production and investment in later quarters without directly changing target output. Differences across province-sector cells reflect transmission through heterogeneous regional technologies, initial conditions and fixed trade relationships, not endogenous network formation or heterogeneity among firms within a cell. National investment, production, employment, prices and fiscal flows aggregate these decentralized and potentially rationed paths. Section 3 assesses the model’s baseline behaviour before the policy experiment is introduced in Section 4.

3 Validation Strategy and Empirical Performance

3.1 Model purpose, domain and validation strategy

MacroABM-CA is designed to study how firm- and household-level decisions propagate through provincial production structures and aggregate into Canadian macroeconomic outcomes. Its empirical domain is therefore broader than forecasting a single national series: credibility depends on the consistency of initialized agent states, the operation of behavioural and market mechanisms, the resulting province-sector transactions, and the aggregate time paths generated by their interaction.

Following Tieleman (2022), validation is organized relative to this purpose and across the model’s micro, meso and macro levels. Lower-level mechanisms provide inputs to higher-level mechanisms, whose joint implications can generally be assessed only through simulation. The strategy consequently distinguishes **mechanisms**, such as adaptive demand formation, input rationing and household consumption out of disposable income, from **targets**, such as historical GVA, inflation and unemployment rates. Matching an aggregate target does not by itself validate the underlying mechanism; conversely, a well-grounded initialization does not establish that every behavioural parameter has been empirically estimated (Fagiolo, Moneta, and Windrum 2007; Tieleman 2022).

Agent states and behavioural rules

|

v

Market interaction and realization

|

v

Province-sector transactions and constraints

|

v

National production, employment, prices and fiscal outcomes

Realized outcomes update incomes, stocks and expectations at each level.

This hierarchy matches the model’s intended contribution: linking behaviour to realized transactions and then to regional and national outcomes. It also bounds the validation claim. MacroABM-CA combines empirically constructed initial conditions, inherited behavioural parameterization and targeted mechanism checks; not every agent rule is directly estimated from Canadian microdata.

3.2 Micro- and meso-level evidence

At the micro level, the principal evidence concerns initialization and internal consistency. Province-sector production, employment, capital stocks and compensation are mapped to the common 50-sector classification using Statistics Canada and OECD sources. The synthetic household population is scaled to 15.455 million households and reproduces selected Canadian controls for tenure, income and balance-sheet distributions; for example, the share of owner-occupied households is controlled to the 65.4 percent reported for 2022 by Statistics Canada’s Canadian Housing Survey. The household construction preserves the joint dependence among financial variables rather than matching only their separate marginals. These checks support the initial distribution of agent states.

At the meso level, accounting checks verify that the 2022 interprovincial input-output system balances and that transactions retain their sector, origin and destination dimensions. Controlled perturbations also verify the intended ordering of key mechanisms: changes in planned purchases affect realized transactions and stocks, which subsequently affect production and demand. Numerical checks require finite values, non-negative physical stocks and reproducibility under common random-number streams. These are necessary implementation and mechanism tests, not direct empirical validation of Canadian rationing frequencies or supplier-level responses.

The observed IO network and its technical coefficients are fixed during the present experiments. Endogenous outcomes include quantities transacted over that structure, unmet orders, prices, stocks, incomes and subsequent plans. The analysis can therefore assess heterogeneous transmission across the 650 province-sector cells, but it does not identify endogenous network formation or adjustment within the distribution of firms represented by a cell.

3.3 Historical macroeconomic validation

Historical validation compares the ensemble-average national model path with Statistics Canada outcomes through 2024. The reported historical specification uses $\alpha = 0.40$ and the four labour and input-planning changes described above. As a deliberately parsimonious external benchmark, we also estimate univariate AR(1) models in the spirit of the forecasting comparisons in Poledna et al. (2023) and Wiese et al. (2024). Their parameters use only observations through 2021Q4, after which the models are projected recursively to 2024Q4 from a fixed origin without updating from realized evaluation-period observations. The exercise remains a conditional historical simulation rather than a pure out-of-sample forecast because observed provincial demand and labour-force paths enter the MacroABM-CA historical environment. The AR(1) is intended as a minimum benchmark: a transparent, univariate reference point that any structural model should approach. It is not the most demanding comparison available. Because MacroABM-CA generates output, unemployment, prices and interest rates jointly, a multivariate vector autoregression (VAR) forecasting these variables together is the more natural statistical counterpart. Ideally that VAR would be conditioned on the same observed exogenous paths, so that both models use the same information set. Such comparisons, in the spirit of the VAR and DSGE benchmarks of Poledna et al. (2023), are part of the ongoing validation work described in Section 8. Fit is summarized by the root-mean-square error (RMSE) and mean absolute error (MAE), both in the units of each series; lower values indicate a closer match to the data.

Variable	MacroABM-CA RMSE	MacroABM-CA MAE	AR(1) RMSE	AR(1) MAE
Real GVA growth (percentage points per quarter)	0.4360	0.3635	0.3542	0.2941
Unemployment rate (percentage points)	0.9219	0.7997	1.3392	1.2781

Table 2. National historical fit against a fixed-origin AR(1) benchmark, 2022-2024. Appendix Table A4 reports the same comparison with the benchmark construction and scoring windows.

Notes: unemployment errors cover 2022Q1-2024Q4. Real-output-growth errors cover 2022Q2-2024Q4 because the saved MacroABM-CA path does not contain a 2021Q4 level from which to calculate model growth in 2022Q1. MacroABM-CA is a conditional historical simulation, so this comparison measures historical fit rather than establishing a controlled out-of-sample forecasting ranking.

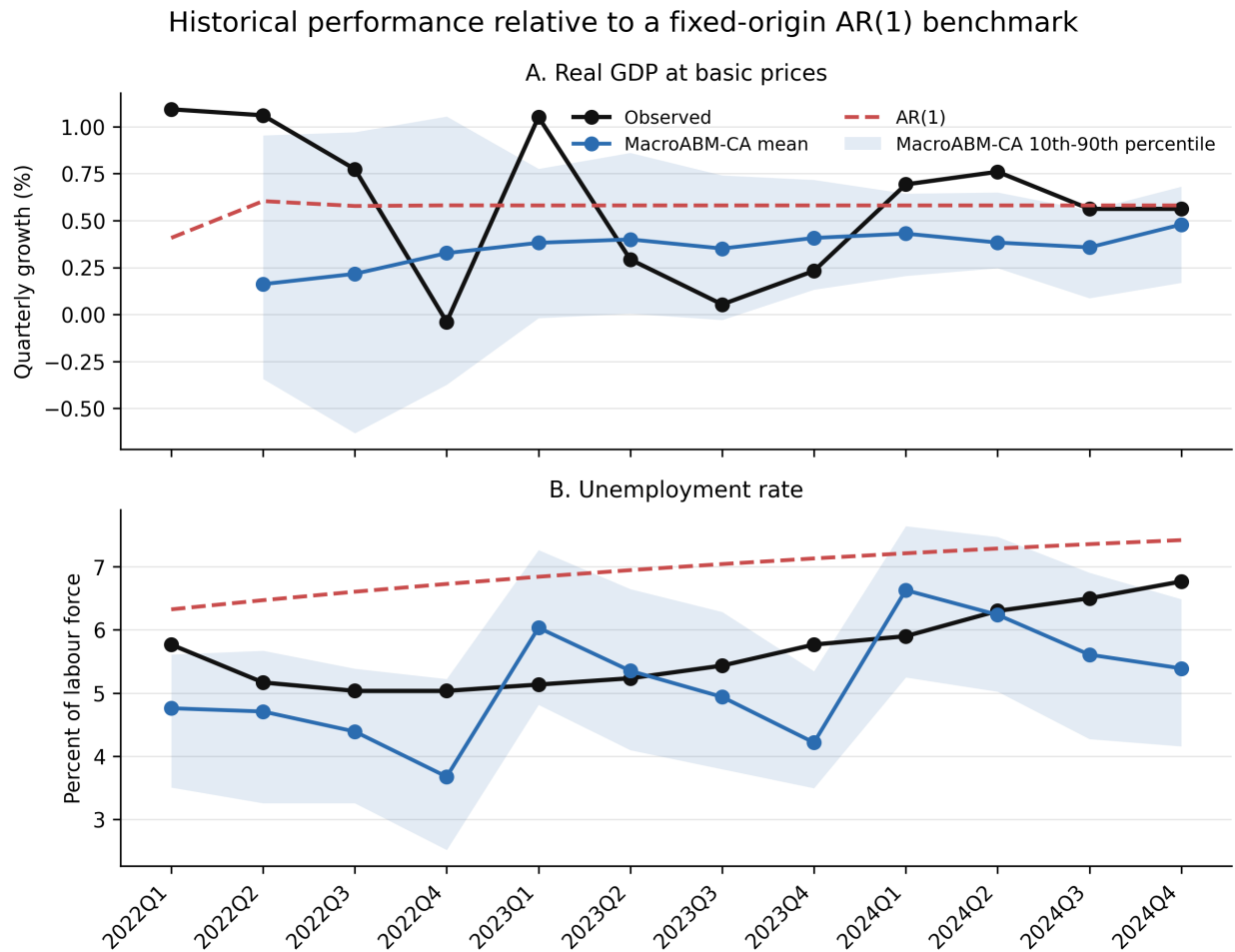


Figure 2. Historical national performance relative to a fixed-origin AR(1) benchmark, 2022-2024. Panel A compares quarterly growth in Statistics Canada real GDP at basic prices with growth in MacroABM-CA aggregate real GVA and the AR(1) forecast. Panel B compares quarterly-average unemployment rates. Lines show observed outcomes, the MacroABM-CA simulation average and the recursive AR(1) forecast; shading denotes the 10th-to-90th percentile range across MacroABM-CA simulations. The two approaches use different information sets, as discussed in the text.

MacroABM-CA’s GVA errors are 23.1 percent larger for RMSE and 23.6 percent larger for MAE than those of the AR(1), a competitive result given the different information structures and short evaluation window. For unemployment, MacroABM-CA’s errors are 31.2 and 37.4 percent smaller. The national results are stronger than the provincial scorecard partly because aggregation gives greater weight to large provinces and permits offsetting provincial errors to cancel. At the provincial level, simulated GVA growth is less dispersed than observed growth. Unemployment errors are more pronounced in several small provinces, where coarse synthetic-population resolution and common behavioural parameters have greater influence on equal-weight error measures. These findings indicate both aggregation effects and structural limitations in representing province-specific labour markets; the national fit should not be interpreted as uniform regional validation. Appendix C.1 reports the provincial scorecard behind this diagnosis (Figure A3), and Appendix C.2 documents the data, estimation sample and coefficients of the AR(1) benchmark.

3.4 Scope of validation and remaining limitations

The validation evidence is strongest for data construction, accounting consistency, qualitative mechanism operation and national labour-market dynamics. It is weaker for province-level dynamics, realized bilateral transactions and behavioural parameters that have not been estimated specifically for Canada. Provincial errors reflect compressed cross-province dispersion, common behavioural parameters and coarse population resolution in small provinces. Population-weighted national rates and treatment-control changes are consequently more informative than equal-weight provincial averages or unadjusted agent counts.

The policy experiments use the endogenous household, fiscal and price specification described in Section 2 from initialization. This environment is not presented as a historical backcast: historical validation and forward policy analysis serve different purposes. Validation asks whether the model reproduces observed Canadian history; policy analysis compares two simulated futures that differ only in whether the policy is applied. Because errors common to both futures cancel in the comparison, a policy effect can be informative even where the model’s absolute path is imperfect, but only to the extent that the mechanisms driving the difference are credible. Matched treatment-control comparisons remove common stochastic and baseline movements, not uncertainty about behavioural elasticities, the synthetic fiscal perimeter or the fixed production network. These limitations motivate the sensitivity and mechanism decompositions reported in Section 5 and are discussed together in Section 7.

4 Canadian Policy Context and Experimental Design

4.1 Canadian clean-economy investment tax credits

Beginning in 2021, the federal government announced a suite of refundable Clean Economy Investment Tax Credits. Five credits are now legislated and claimable: the Carbon Capture, Utilization and Storage, Clean Technology, Clean Hydrogen, Clean Technology Manufacturing and Clean Elec-

tricity ITCs (Government of Canada 2026a). Their announcement dates, eligible expenditures, rates and implementation dates differ, so the suite should not be treated as a single uniform programme.

The Clean Technology ITC provides the most transparent benchmark for the present application because it operates directly through eligible capital expenditure. It offers a refundable credit of up to 30 percent for qualifying new clean-technology property acquired and available for use from March 28, 2023 through 2033, falling to 15 percent in 2034 and ending thereafter (Government of Canada 2026b). This design supplies a direct firm-level incentive whose propagation through investment, suppliers, employment, incomes and demand can be observed in MacroABM-CA.

4.2 Policy question and experimental scope

The experiment asks how a temporary reduction in the effective user cost of eligible capital propagates through Canada’s regional economy. It implements a stylized investment tax credit inspired by the design of Canada’s Clean Technology ITC, not a statutory microsimulation, an ex post programme evaluation or a forecast of actual claims. Eligible assets are proxied by purchases of electrical equipment (C27) and machinery and equipment n.e.c. (C28). These broad producing sectors neither isolate clean technologies nor reproduce legal eligibility conditions.

The simulated credit rate is 30 percent and the intervention operates from the first quarter of 2026 through the fourth quarter of 2030. The model does not directly change firms’ target production, productivity, credit supply, labour supply or technical coefficients. The experiment isolates one transparent behavioural premise: firms with greater eligible exposure expand their desired capital programme more strongly when its effective cost falls.

4.3 Behavioural implementation

For firm (i), let the lagged eligible share be the value share of C27 and C28 in the preceding planned capital bundle, where $(I^*_{ij,t-1})$ denotes the firm’s planned desired capital order for input (j) defined in Section 2.5:

$$s_{i,t-1} = \frac{\sum_{j \in E} p_{j,t-1} I^*_{ij,t-1}}{\sum_j p_{j,t-1} I^*_{ij,t-1}}, \quad E = \{C27, C28\}. \quad (8)$$

During the policy window, the credit scales the capital requirement by

$$m_{i,t} = (1 - \tau s_{i,t-1})^{-\eta_I}, \quad \tau = 0.30, \quad \eta_I = 0.50, \quad (9)$$

with $(m_{i,t}=1)$ otherwise. When a firm’s planned capital bundle is empty, so the denominator in equation (8) vanishes, the eligible share is set to zero and $(m_{i,t}=1)$. The multiplier enters before the model compares the desired capital requirement with existing stocks. It therefore raises desired gross investment while preserving the baseline link between expected production, capital requirements and stock adjustment. Applying the multiplier to the project-wide bundle represents complementarity between eligible machinery and other capital required to undertake the associated project. The elasticity is externally specified and should be interpreted as the responsiveness of the desired capital programme to the eligible-share-weighted user-cost reduction, not as a parameter estimated within MacroABM-CA.¹

1. The elasticity is imposed rather than recovered endogenously because the inherited investment block offers no

The economic sequence is

eligible exposure and ITC

- > lower effective project user cost
- > higher desired capital requirement
- > stock adjustment and desired orders
- > financing and goods-market realization
- > installed capital and production
- > income, sales and adaptive demand
- > subsequent production and investment plans.

Appendix D.1 provides the eligible-share definition, the zero-eligible-share treatment, the pre-stock ordering of the multiplier, the refund calculation and the timing of purchase, installation and later productive use.

4.4 Counterfactual and outcome measurement

Each treatment simulation is paired with a no-ITC control using the same initial state, behavioural configuration and random-number stream. Results are reported as averages across matched simulations. The main monetary environment holds the policy rate constant, while the robustness analysis allows it to respond through the Taylor-type rule described in Section 2.9. Annual flows are calendar-year sums of quarterly values; stocks are measured at year-end; rates are annual averages or year-end values as stated.

Gross ITC refunds are measured as 30 percent of realized treatment purchases of eligible C27 and C28 assets during 2026-2030. They therefore include inframarginal purchases and are distinct from incremental investment, and they are a gross outlay rather than the net fiscal cost, which also depends on the offsetting changes below. Refunds are zero after 2030. Changes in represented taxes, cash benefits, government purchases and debt interest are reported separately. Because the synthetic regional governments combine fiscal instruments with different institutional perimeters, absolute deficit and debt paths are diagnostic rather than forecasts of Canadian public finances.

5 Results

5.1 Aggregate magnitude of the response

The ITC raises desired and realized investment by an average of \$36.82 billion during 2026-2030 in constant prices, about 1.6 percent above the no-policy (control) path, with the end-2030 capital stock \$29.01 billion above control at current prices. Over the same period, cumulative real production is \$103.91 billion higher (0.39 percent above control) and real GVA is \$49.33 billion higher

internal margin through which the credit could act. Desired capital follows a demand-led accelerator, combining depreciation-driven replacement with partial adjustment of the stock toward a production-based reference, and contains no user-cost or capital-price term (Wiese et al. 2024); under the Leontief technology a cheaper capital good also induces no substitution toward capital. The two more neutral alternatives are consequently weak in this configuration. Booking the refund purely as additional firm cash flow would operate only through the credit channel, which transmits weakly here, and would therefore leave investment almost unchanged. Routing it through productivity would change the character of the policy. We therefore introduce an explicit user-cost wedge governed by η_I , the one transparent and tunable channel that produces a non-trivial investment response. Making that wedge visible and reporting its sensitivity (Appendix F.3, Figure A6) is preferable to embedding an equivalent responsiveness implicitly inside an aggregate investment equation. The cost is that the level of the response is assumed rather than identified, so the model's contribution is the propagation of the impulse rather than its size.

(0.35 percent). Employment at the end of 2030 increases by approximately 145,054 persons and the unemployment rate falls by 0.259 percentage points. These are treatment-control effects in the model economy, not forecasts of the actual Canadian programme.

Table 3. Aggregate treatment-control effects of the investment tax credit (simulation averages). These are the paper’s headline results under the constant-rate specification and are the reference point for the central estimates; the monetary and elasticity exercises below are reported separately and are not constructed to reproduce these averages.

Outcome	2026-2030 average effect	2031-2035 average effect	2026-2035 average effect
Desired investment (cum., constant-price \$B)	36.82	45.56	82.38
Realized investment (cum., constant-price \$B)	36.82	45.56	82.38
Installed capital stock value (end-period, current-price \$B)	29.01	n.r.	n.r.
Real production (cum., fixed-price \$B)	103.91	272.49	376.40
Real GVA (cum., fixed-price \$B)	49.33	133.27	182.60
Employment (end, persons)	145,054	194,915	194,915
Unemployment rate (end, pp)	-0.259	-0.724	-0.724
Household consumption (cum., constant-price \$B)	12.99	44.63	57.62
Imports (cum., constant-price \$B)	15.86	41.54	57.40
Gross ITC refunds (cum., constant-price \$B)	50.14	0.00	50.14

Notes: effects are treatment minus control and are averages across matched simulations. All flow quantities are expressed in constant prices: production and GVA at fixed initialization prices, and investment, household consumption, imports and refunds deflated to constant prices by the GVA deflator. The installed capital-stock value is the sole exception, reported at end-period current prices; long-horizon current-price differences are not reported (“n.r.”) because treatment-control price differences materially affect the valuation of the inherited stock. Flows are cumulative within each window; capital and labour-market entries are end-2030 and end-2035 differences. The gross refund shown here in constant prices (\$50.14 billion) is the \$59.31 billion nominal fiscal outlay of Table 5 deflated to constant prices.

Relative to gross refunds during the active window, and with all quantities in constant prices, the mean responses are \$0.73 of additional investment, \$2.07 of additional production and \$0.98 of additional GVA per dollar of credit. Imports rise by \$0.32 per dollar of refunds. These ratios

express constant-price output responses per constant-price dollar of gross refunds (a \$50.14 billion refund, deflated from the \$59.31 billion nominal outlay reported in Table 5).

The production response is about twice the GVA response because the two measures count different things. Production (gross output) records every sale along a supply chain, so inputs are counted each time they change hands: steel sold to a parts maker, then again inside the parts sold to a machinery producer, and again inside the finished machine. GVA records only the value each firm adds, its sales less the inputs it buys, and so counts each dollar of final value once; it is the measure that corresponds to GDP. The ratio of the two responses, about 2.1, is close to the ratio for the Canadian economy as a whole, where 2022 gross output of \$5,029 billion is about 1.9 times GVA of \$2,674 billion (Appendix B). It is slightly higher because the credit directs demand toward machinery and other manufacturing industries, which purchase a large volume of intermediate inputs relative to the value they add. The larger production figure therefore reflects the depth of the supply chains that the policy activates, not additional economic activity beyond that measured by GVA.

These magnitudes can be read in three ways. First, the credit is modest relative to the economy: refunds average about \$10 billion a year in constant prices, roughly 0.35 percent of GVA, and the GVA response is of the same relative size. Second, investment rises by less than the refund (\$0.73 per dollar) because refunds are paid on all eligible purchases, including the large volume of machinery that firms would have bought anyway; only the increment is new investment. Third, GVA per dollar of refunds is close to one, which is of the same order as the government-spending multipliers of roughly 0.6 to 1 surveyed by Ramey (2019). That comparison is only indicative, however. A fiscal multiplier measures the output response to a marginal change in public spending under a specified financing rule, whereas our ratio divides an average response by a gross refund that includes inframarginal purchases and is deficit-financed. The ratios are therefore descriptive response measures, not structural fiscal multipliers or benefit-cost estimates, and they should not be compared one-for-one with such estimates.

Imports rise alongside investment and production rather than falling, for two reasons. The eligible machinery sectors (C27 and C28) are net importers in the 2022 data, so a large share of the additional equipment demand is met from abroad (Appendix B.2); and the higher employment and income generated by the policy raise household spending, part of which is also imported. The credit lowers the cost of equipment regardless of where it is produced, so it does not by itself favour domestic suppliers. Section 5.6 discusses this foreign-sourcing response further.

The effects over 2031-2035 exceed those over 2026-2030 even though no refunds are paid after 2030. This reflects persistence rather than a new impulse: the active-window totals include years in which the response is still building up, while the post-policy totals start from the higher level reached by 2030 and are sustained by firms' updated demand expectations. Sections 5.2 and 5.3 explain this mechanism.

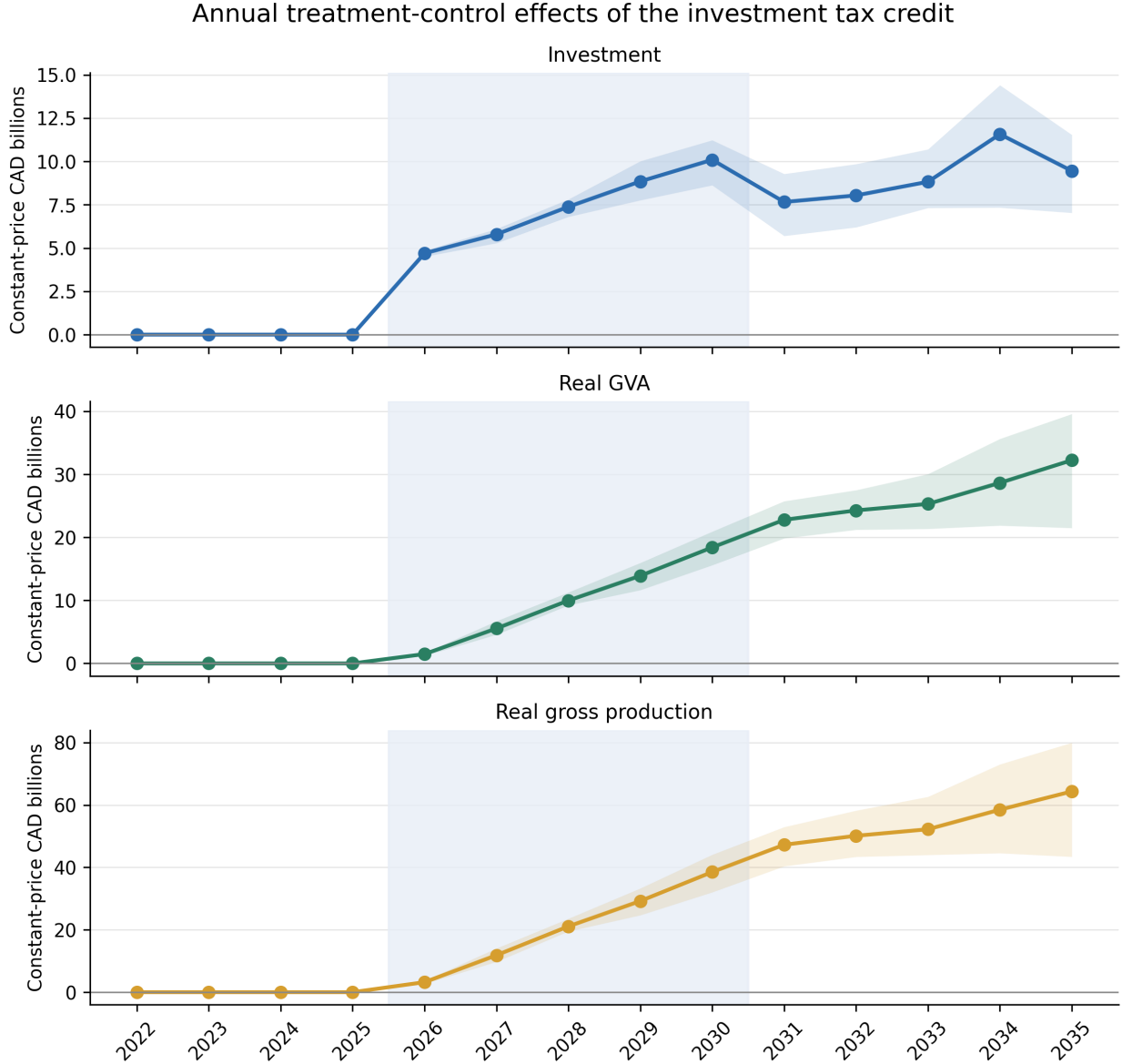


Figure 3. Annual treatment-control effects of the investment tax credit. All series are in constant prices: investment is deflated to constant prices by the GVA deflator, and real GVA and real gross production are valued at fixed initialization prices. Lines show simulation averages and shading denotes the 10th-to-90th percentile range. The shaded vertical interval marks the 2026-2030 policy window.

5.2 Dynamic propagation and persistence

At the aggregate level, desired and realized investment move together, indicating that capital orders are not materially rationed in these simulations. The distinction remains economically important because other desired inputs and outputs need not be realized, and the investment response alters stocks and demand over time. The same-state diagnostic (Appendix D.2) separates the contemporaneous increment generated by the user-cost multiplier from differences inherited through earlier policy-induced changes in stocks, prices and demand histories.

The response persists after the credit expires. During 2031-2035, refunds are exactly zero, yet cumulative investment remains \$45.56 billion above control in constant prices, production \$272.49 billion higher and GVA \$133.27 billion higher. The long-horizon current-price capital-stock difference is not reported, because after the policy window treatment-control price differences materially affect the valuation of the inherited stock (Appendix F.1). This continuation arises from realized income and sales, evolving stocks and the updating of subsequent plans. We therefore describe the result as finite-horizon persistence, not permanent hysteresis or a steady-state effect.

Two qualifications guard against over-reading the post-policy totals. First, the two windows are not comparable ramp stages: the active-window cumulative is dominated by the years in which the policy is still building from zero, whereas the post-policy cumulative starts from the elevated end-2030 level, so a post-policy total that exceeds the active-window total partly reflects a sustained level rather than a new impulse after expiry. The annual GVA effect rises from \$1.5 billion in 2026 to \$18.4 billion in 2030; had it merely stayed at its 2030 level, cumulative GVA over 2031-2035 would be about \$92 billion, already well above the \$49.33 billion of the policy window. The actual total, \$133.27 billion, is larger still because the effect continues to grow after expiry, reaching \$32.2 billion in 2035, or 1.08 percent of control GVA. This further growth comes almost entirely through the adaptive-demand channel of Section 5.3, whose annual contribution rises from \$12.0 billion in 2030 to \$29.7 billion in 2035, while the fixed-expectations contribution fades. Within the simulated horizon the model therefore shows no reversal toward the no-policy path, because nothing in the maintained specification acts to pull expectations back once higher sales have been realized. Second, the persistence operates through installed capital and, more importantly, adaptive expectations: firms that experienced higher realized demand during the policy carry that memory forward under the reported adjustment speed, sustaining production and input plans after refunds end. The magnitude of this tail therefore depends on the adaptive-expectations speed and the input-planning assumptions of Section 2.5, which are disciplined by historical fit rather than estimated structurally. The sensitivity analysis in Section 5.7 varies the investment elasticity and the monetary rule but not these expectation and planning assumptions, so the post-policy magnitudes should be read as indicative of persistence under the maintained specification rather than as precise long-horizon estimates.

5.3 Adaptive-demand mechanism

Why does a temporary credit keep affecting the economy after it ends? The main reason is that firms learn from experience. During the policy, firms sell more, and because their demand expectations are based on recent sales (Section 2.4), they expect higher demand in later quarters and plan more production and investment accordingly. These plans generate more income and sales, which reinforce the higher expectations. Realized outcomes therefore change the information on which later plans are based, and this feedback continues after the refunds stop.

To measure how much of the response comes from this feedback, we run an additional simulation in which the policy is active but firms' demand expectations cannot respond to it: each firm's record of its own past sales is replaced by the record from the no-policy simulation, while the credit, current purchases, household behaviour, prices and all other feedbacks operate as usual. Comparing the three simulations splits the total policy effect into two parts:

- **Channel A, the fixed-expectations channel**, is the effect of the policy when firms' demand expectations are held at their no-policy values. It captures the investment increase itself, the resulting purchases and capital, and every other feedback that does not work

through firms' sales expectations. It is broader than a pure direct or physical effect.

- **Channel B, the adaptive-demand channel**, is the remainder: the total policy effect minus Channel A. It measures the additional response that arises because firms update their demand expectations in light of policy-induced sales, including the interaction of that updating with other mechanisms.

Controlled-path counterfactuals of this kind, in which one mechanism is held to a reference path while the rest of the model remains active, are standard in quantitative macroeconomics and are consistent with Tieleman's (2022) argument that mechanisms generated by interacting agent rules must be revealed through simulation. The split depends on which mechanism is held fixed and should not be interpreted as a unique structural or causal partition. Appendix D.3 gives the formal definition of the two channels, and Table A5 repeats the decomposition.

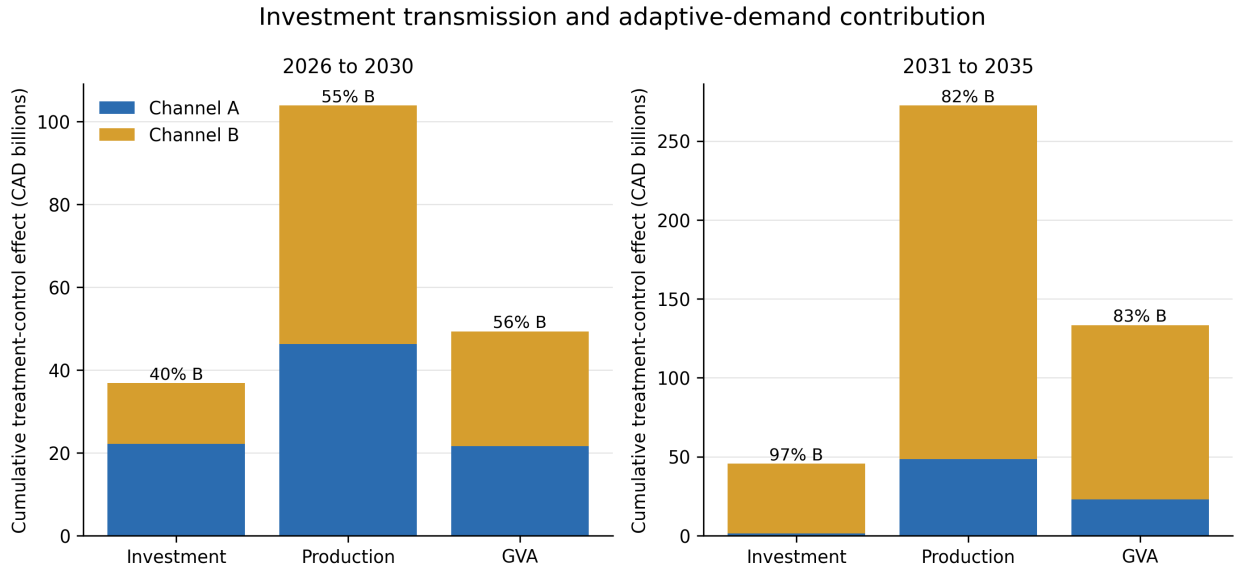


Figure 4. Decomposition of the policy response into the fixed-expectations channel (Channel A, firms' demand expectations held at their no-policy values) and the adaptive-demand channel (Channel B, the additional response from firms updating their expectations). All series are in constant prices: investment is deflated by the GVA deflator, and production and GVA are valued at fixed initialization prices. Channel B shares are ratios of the average Channel B response to the average full response.

Table 4. Decomposition of the policy response into the fixed-expectations channel (Channel A) and the adaptive-demand channel (Channel B). Channels are computed on the same headline experiment set as Table 3, so Channel A and Channel B sum to the headline treatment-control averages. All quantities are in constant prices.

Cumulative response (constant-price \$B)	Channel A: fixed expectations	Channel B: adaptive demand	Channel B share (%)
Investment, 2026-2030	22.11	14.71	40.0
Production, 2026-2030	46.30	57.60	55.4
GVA, 2026-2030	21.67	27.66	56.1
Investment, 2031-2035	1.33	44.22	97.1
Production, 2031-2035	48.41	224.08	82.2
GVA, 2031-2035	22.80	110.47	82.9

Notes: shares are calculated as the average Channel B response divided by the average full response. All quantities are in constant prices, following Table 3.

The adaptive-demand channel accounts for slightly more than half of the production and GVA responses during the policy window and more than four-fifths after it (Table 4). After 2030, almost all of the additional investment (97 percent) comes through this channel: once refunds stop, firms keep investing more mainly because they expect higher demand, not because the credit is still lowering their costs. This does not mean that capital accumulation is unimportant, because the adaptive response develops in an economy whose capital stocks and transactions have already been changed by the policy.

5.4 Regional and sectoral heterogeneity

The province-sector analysis distinguishes absolute contributions from responses normalized by each cell's control baseline. Cells differ in scale, eligible-capital exposure, production coefficients, import dependence, initial stocks and position in the fixed interprovincial IO structure. These dimensions provide the basis for assessing how similar investment impulses are converted into realized production across regions and sectors.

Across cells with stable pre-policy denominators, the median cumulative production response is 0.58 percent of pre-policy production scale, the mean is 1.24 percent and the 10th-to-90th percentile range is 0.08 to 3.38 percent. This dispersion is economically meaningful. The mean is more than twice the median, so the distribution has a long right tail: most province-sector cells respond modestly, while a minority respond several times more strongly. The policy's effects are therefore concentrated in particular industries and places rather than spread evenly, which is precisely the kind of information that an aggregate model cannot provide. The correlation between the normalized direct investment impulse and the normalized production response is 0.35, so a cell's own investment exposure explains only part of its production response; the rest depends on its position in supply chains.

Median responses are highest in several machinery, electronics, fabricated-metal, wood-product and transport-equipment industries, including the two eligible sectors C27 and C28. Three features plausibly explain this pattern. First, C27 and C28 produce the eligible equipment, so they face the most direct increase in demand. Second, electronics and fabricated-metal products are important suppliers of components to machinery and electrical-equipment production, so they receive the next round of intermediate demand through the input-output network. Third, wood products and transport equipment are relatively capital-intensive industries with large machinery shares in their capital bundles, so a lower cost of eligible machinery raises their own desired investment and, through the adaptive-demand channel, their production. These explanations are consistent with the model structure but remain descriptive: the comparisons do not hold other structural differences constant.

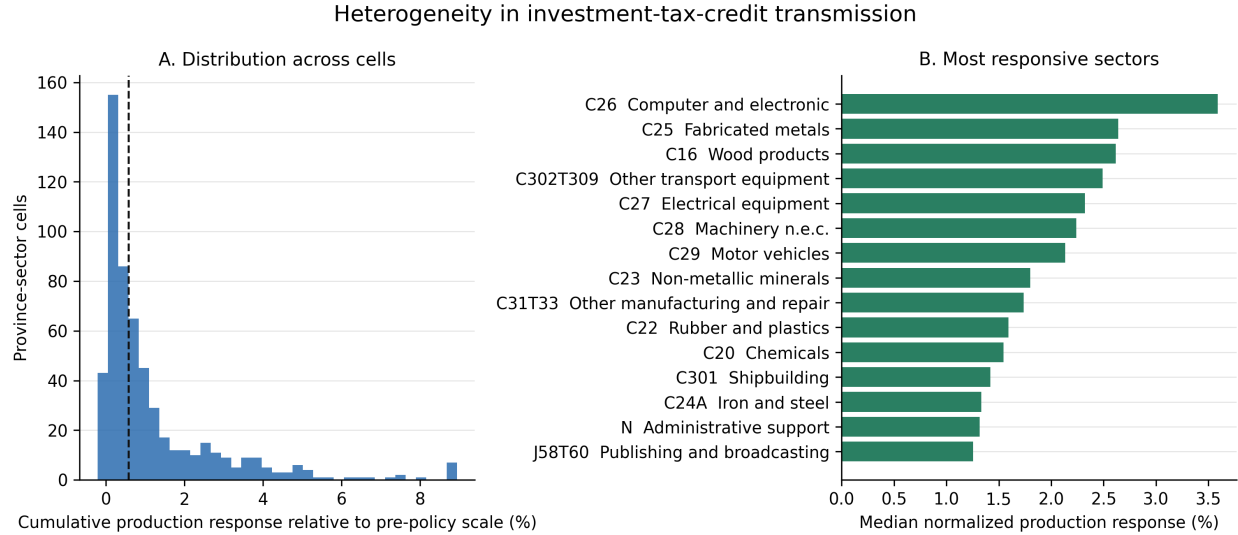


Figure 5. *Heterogeneity in investment-tax-credit transmission across province-sector cells.* Panel A reports the cumulative 2026-2030 real-production response relative to twenty quarters of average pre-policy production; the displayed distribution is winsorized at the first and ninety-ninth percentiles, and the dashed line marks the median. Panel B reports median normalized responses for the fifteen most responsive sectors. Cells with near-zero pre-policy production are excluded from ratio calculations.

The model traces propagation over a fixed production and trade network; it does not model endogenous supplier formation. Lerche’s (2025) distinction between own and IO-linked responses motivates the descriptive analysis, but MacroABM-CA does not reproduce his empirical identification or establish causal network spillovers. The analysis here therefore reports own-exposure heterogeneity and leaves first-order customer- and supplier-exposure measures defined over the fixed input-output structure to future work. Appendix E.2 (Figure A4) reports how the normalized response of each province-sector cell relates to its reliance on international imports and on suppliers in other provinces: responses are modestly higher where import shares are higher and essentially unrelated to interprovincial sourcing.

The response can also be traced at the provincial level. Ontario, Alberta and Quebec receive 73.3 percent of gross refunds and generate 75.6 percent of the additional national GVA, a concentration that largely reflects their economic scale and eligible investment base. Ontario has the largest absolute response, with \$12.04 billion of additional investment, \$37.13 billion of additional production, \$16.53 billion of additional GVA and about 51,700 additional employed persons. Normalized outcomes differ from this ranking: among the larger provinces, Saskatchewan shows the largest GVA response relative to its pre-policy scale, about 0.60 percent, and the largest decline in unemployment, about 1.34 percentage points. Figure 6 decomposes each provincial response into Channel A and Channel B; Channel B contributes roughly 52 to 64 percent of the provincial GVA responses. Adaptive propagation is therefore geographically widespread rather than confined to the provinces that receive most of the refunds, and the national response is not driven by a single province. These are descriptive treatment-control differences across provinces and do not identify causal spillovers between them.

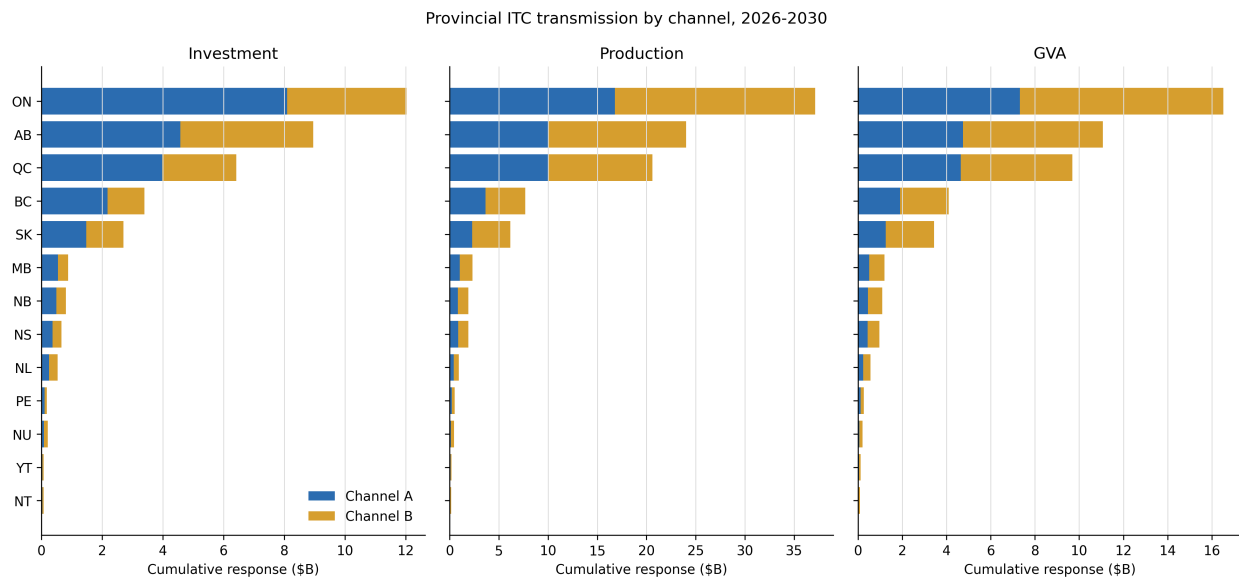


Figure 6. Provincial decomposition of the 2026-2030 investment-tax-credit response into the fixed-expectations channel (Channel A) and the adaptive-demand channel (Channel B), for investment, production and gross value added. Bars show treatment-control simulation averages by province, stacked into Channel A and Channel B contributions, in constant prices. The full provincial scorecard, including employment, unemployment and imports by province, appears in Table A6 (Appendix E.1).

5.5 Monetary-policy environment

The main experiment holds the policy rate constant. A robustness exercise instead endogenizes the common rate through the stylized Taylor-type rule while retaining the same household, fiscal, banking and investment architecture. The common rate then passes into commercial-bank lending rates, credit outcomes, cash flows and demand. During 2026-2030, the treatment-control response of the policy rate averages -0.057 percentage points under the Taylor environment, and the corresponding firm lending-rate response averages -0.035 percentage points. These small differences indicate limited direct monetary opposition to the investment incentive in this exercise.

The channel decomposition of Section 5.3 is more informative than a comparison of raw aggregate mean responses. Production through the fixed-expectations channel (Channel A) is \$43.13 billion under the Taylor rule and \$43.80 billion under the constant rate, while Channel A GVA is \$20.31 billion and \$20.62 billion, respectively. Thus the investment mechanism remains operative and its direct production and value-added transmission is essentially unchanged across the two monetary environments. The remaining difference between the aggregate responses arises through the adaptive-demand channel (Channel B), where monetary feedback interacts with sales, demand histories and subsequent plans. The constant-rate figures reported here are drawn from the five seeds (0 to 4) for which the Taylor exercise was run, so they are a five-seed matched subset of the constant-rate runs, whereas Table 4 reports the ten-seed ensemble mean; the Taylor rule was not run for the remaining seeds. The relevant comparison is therefore between the two monetary environments within this matched five-seed exercise, not against the ten-seed headline decomposition. This is a stylized robustness exercise rather than a systematic estimate of monetary amplification or attenuation, or a calibrated Bank of Canada experiment. Figure A5 in Appendix F.2 shows the comparison by channel.

5.6 Imports and fiscal incidence

Imports rise by \$15.86 billion during 2026-2030 and by a further \$41.54 billion during 2031-2035, in constant prices. These totals include the foreign content of capital purchases and imports induced through intermediate and household demand. They should be interpreted as the foreign-sourcing response associated with the policy, not as a formally identified leakage rate.

This foreign-sourcing response is one of the clearest structural lessons of the experiment. It follows directly from the 2022 economy on which the model is built: both eligible sectors are net importers, with international import use of \$33.9 billion for electrical equipment (C27) and \$64.6 billion for machinery n.e.c. (C28) (Appendix B.2). An incentive that raises demand for these goods therefore draws in imports as well as domestic production. Because the production network is fixed, the model does not allow the credit to build new domestic capacity in these industries, so the import response should be read as the leakage implied by Canada’s existing supply structure.

The gross refund is a nominal fiscal outlay of \$59.31 billion during the policy window and zero thereafter, equal to the statutory rate applied to all treatment purchases of eligible assets, including purchases that would also have occurred in the control; expressed in constant prices it is \$50.14 billion. The total project-wide investment response is \$36.82 billion in constant prices because the behavioural rule also expands complementary non-eligible capital and because later model states and demand affect investment. Represented tax, benefit, purchase and interest changes are distinct from gross refunds and are reported at current prices in Table 5. Absolute deficit and debt levels remain diagnostic because the synthetic government perimeter and absence of an explicit government-bond holder limit their economic interpretation.

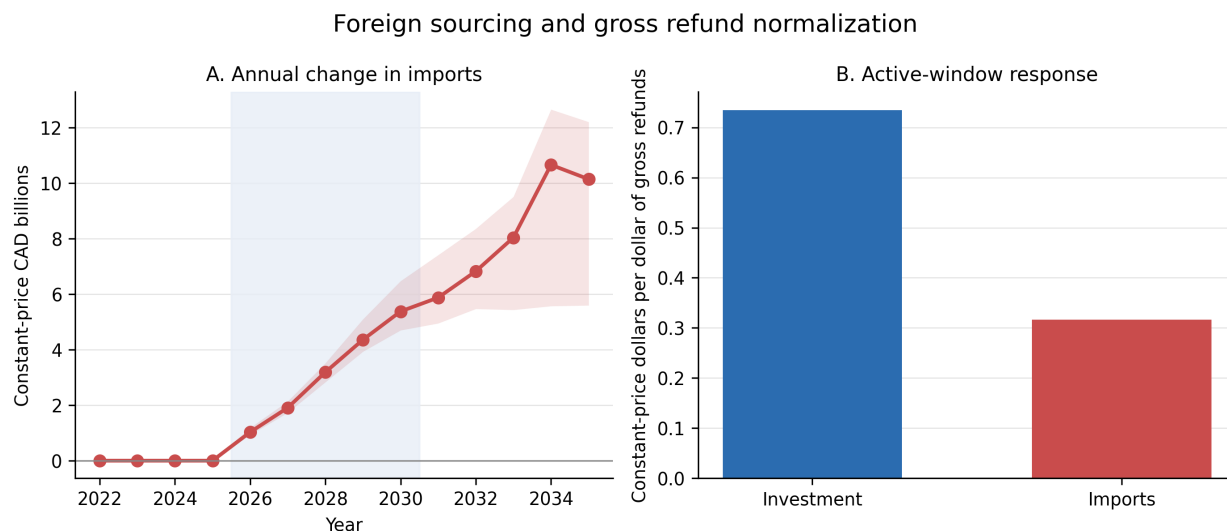


Figure 7. Foreign sourcing and gross refund normalization. Panel A reports annual treatment-control changes in imports in constant prices; the line is the simulation average, the band is the 10th-to-90th percentile range and the shaded interval marks the policy window. Panel B reports active-window additional investment and imports per dollar of gross refunds. Both ratios compare constant-price quantities and are descriptive response measures, not fiscal multipliers.

Table 5 reports the represented fiscal flows separately. In the model the credit is deficit-financed: government purchases are exogenous and no offsetting tax increase, spending cut or fiscal reaction rule accompanies the refund, so the outlay adds directly to the deficit and to debt and there

is no financing-side drag on household consumption. The gross outlay is nonetheless partly self-offsetting within the fiscal perimeter. During the policy window represented tax revenue rises by \$8.26 billion, led by corporate and personal income tax, and cash benefits fall by \$3.70 billion as employment increases, so the deficit rises by about \$48.9 billion, roughly four-fifths of the \$59.31 billion gross refund. After 2030 no refunds are paid, yet the persistently higher activity still generates about \$29.8 billion of additional represented revenue, so over 2031-2035 the deficit is lower than under control and the tax base recoups a substantial share of the earlier cost. These figures describe fiscal incidence within the model rather than a benefit-cost estimate: they depend on the same assumptions as the output response, and the model represents neither a financing choice nor crowding-out, so a tax-financed or spending-offset version of the same credit would produce smaller demand effects.

Table 5. Represented fiscal-flow treatment-control changes (simulation averages, cumulative, current-price \$B). Positive values raise the corresponding flow.

Fiscal flow	2026-2030	2031-2035
Gross ITC refunds	59.31	0.00
Represented tax revenue	8.26	29.84
of which corporate income tax	5.75	14.84
of which personal income tax	1.74	4.74
Cash benefits and transfers	-3.70	0.59
Government realized purchases	-0.32	8.13
Government interest expense	1.82	2.74

Notes: treatment minus control, averaged across matched simulations. Represented tax revenue is the sum of the taxes represented in the model and need not equal the listed components, which are shown for their relative magnitude. Government realized purchases and the deficit are highly dispersed across simulations, meaning that their values differ widely from one simulation to another, so their averages are imprecise and should not be read as point estimates; the deficit and debt levels are diagnostic given the synthetic fiscal perimeter and the absence of an explicit government-bond holder. Unlike the constant-price results in Tables 3, 4 and 6, this fiscal table is reported at current prices; the \$59.31 billion gross refund here is the same outlay shown in constant prices as \$50.14 billion in Table 3.

5.7 Robustness and interpretation boundaries

The principal active-window investment, production and GVA effects remain positive across the monetary and investment-elasticity exercises. Varying η_I from 0.20 to 1.00 scales the response approximately linearly, with mild concavity at higher values. Active-window investment rises from \$13.22 billion to \$63.04 billion, production from \$35.11 billion to \$167.29 billion and GVA from \$16.92 billion to \$81.85 billion (Table 6). Gross refunds remain nearly flat in constant prices, increasing only from \$46.87 billion to \$47.90 billion because they include the large inframarginal eligible-investment base. The sensitivity therefore changes incremental behaviour much more than the gross fiscal outlay. The exercise holds the remaining model configuration fixed while varying η_I , so it isolates how the response scales with the elasticity. It is run on a single common seed (seed 0), with the control computed once, whereas Table 3 reports the ten-seed ensemble mean; the difference between the $\eta_I = 0.50$ entry here and the Table 3 headline therefore reflects ensemble composition, a single seed versus a ten-seed average, rather than any change in configuration, and the two should not be compared level for level.

Table 6. Sensitivity of the 2026-2030 response to the investment-response elasticity, holding the remaining model configuration fixed. This is a separate sensitivity exercise designed to show how outcomes scale with η_I . Its $\eta_I = 0.50$ entry is a single-seed (seed 0) reference under the headline elasticity, whereas Table 3 is the ten-seed ensemble mean, so the levels should not be compared directly.

η_I	Investment (\$B)	Production (\$B)	GVA (\$B)	Gross refunds (\$B)	Channel B share of GVA (%)
0.20	13.22	35.11	16.92	46.87	49.8
0.35	23.15	62.67	30.44	47.07	53.8
0.50	32.62	87.63	42.53	47.27	55.0
0.75	47.93	129.62	62.62	47.59	57.3
1.00	63.04	167.29	81.85	47.90	59.2

Notes: cumulative treatment-control responses for 2026-2030 under the constant-rate environment. Gross refunds are treatment purchases of eligible assets multiplied by the statutory credit rate. They are not net fiscal costs. Channel B is defined by the controlled-path decomposition in Section 5.3.

Channel B contributes approximately 50 to 60 percent of the GVA response throughout the range, while its share rises moderately with the elasticity. The sensitivity thus changes the scale but not the qualitative sequence of transmission. Longer-horizon effects remain more dispersed and should not be interpreted as steady-state estimates. Figure A6 in Appendix F.3 shows the full sensitivity graphically, including the response per dollar of refunds at each elasticity.

The estimates remain conditional on the externally supplied elasticity, the broad C27-C28 eligibility proxy, the project-wide complementarity assumption and the historically assessed value $\alpha = 0.40$; the experiment also does not incorporate anticipation of the policy or delivery lags for equipment. Section 7 discusses these and the model’s other limitations together.

6 Policy Implications

The experiment is stylized, so its implications concern the design and evaluation of investment incentives rather than the performance of any specific Canadian programme. Five lessons follow from the results.

1. **Gross cost overstates what the credit buys.** Refunds are paid on all eligible purchases, including investment that would have occurred anyway, so each dollar of gross refunds is associated with about \$0.73 of additional investment (Section 5.1). Designs that concentrate support on incremental investment, for example through base-period thresholds or tighter targeting of eligible assets, would raise the investment obtained per dollar of public outlay.
2. **Domestic industrial benefits depend on domestic supply.** The two eligible machinery sectors are net importers in the 2022 data, and imports rise by about \$0.32 per dollar of refunds (Section 5.6 and Appendix B.2). If a policy objective is domestic production and employment rather than investment alone, demand-side credits may need to be paired with supply-side measures, such as the Clean Technology Manufacturing ITC, that expand Canadian capacity to produce the eligible equipment.
3. **Federal incentives have uneven regional effects.** Ontario, Alberta and Quebec receive about three-quarters of refunds and generate about three-quarters of the additional GVA,

mainly because of their scale, whereas several smaller provinces respond more strongly relative to the size of their economies (Section 5.4 and Appendix E). Federal and provincial governments can use this kind of regional profile to anticipate where support will concentrate and to coordinate complementary provincial measures.

4. **Temporary credits have lasting effects, so evaluation should look beyond the eligibility period.** Most of the post-2030 response comes from firms adapting their expectations to the activity the credit generated (Section 5.3). Evaluations that stop when the credit ends will understate its effect, and the timing of phase-outs matters because firms carry the memory of higher demand forward.
5. **Part of the fiscal cost returns through a wider tax base.** Higher activity raises represented tax revenue and lowers benefit payments, offsetting nearly one-fifth of the gross refund during the policy window and producing additional revenue after it (Section 5.6). Net fiscal cost is therefore materially below gross outlay, although the size of this offset depends on how the credit is financed.

Each lesson is conditional on the maintained assumptions discussed in Section 7. The relative patterns (leakage through imported equipment, regional concentration and persistence through expectations) are more robust than the absolute magnitudes, which scale with the assumed investment elasticity.

7 Limitations

The results rest on several modelling assumptions. We state each, its implication for the results and how readers should use the results in light of it.

Imposed behavioural response. The strength of firms’ response to a lower user cost is set by an externally specified elasticity rather than estimated within the model (Section 4.3). The magnitudes in Section 5 therefore scale with this assumption; the sensitivity analysis in Section 5.7 and Appendix F.3 shows that responses scale nearly linearly with it. The results are most informative about how an assumed response is transmitted and distributed, and least informative about the absolute size of that response.

Broad eligibility proxy and no emissions accounting. Eligible assets are represented by two broad machinery sectors (C27 and C28) that do not isolate clean technologies or reproduce legal eligibility rules, and the model does not measure emissions. The experiment should not be read as an estimate of the Clean Technology ITC’s cost or environmental effect.

Fixed production network and technology. Technical coefficients and trading links are fixed at their 2022 values, so firms cannot substitute between inputs, adopt new technologies or form new supplier relationships in response to the policy (Section 2.11). Leakage and regional patterns reflect the 2022 structure and would change if the policy induced new domestic supply.

One representative firm per province-sector cell. Heterogeneity is structural, across provinces and industries, rather than among firms of different size or productivity within an industry (Section 2.1). The model cannot yet show how responses differ between, for example, large and small manufacturers in the same province.

Adaptive expectations and exogenous paths. Firms form expectations adaptively, with a speed chosen for historical fit rather than estimated structurally, and household investment, government purchases and external demand follow exogenous paths. The persistence reported in

Section 5.2 depends on these choices and should be read as indicative rather than as a precise long-horizon estimate.

Stylized fiscal and monetary institutions. Regional governments are synthetic consolidated fiscal units, the credit is deficit-financed and government debt has no explicit bondholder, so absolute deficit and debt levels are diagnostic (Section 2.8). The Taylor-type rule is a transparent stylization rather than an estimated Bank of Canada reaction function (Section 2.9).

Validation scope. The model fits national unemployment better than a simple benchmark but national output growth less well, and provincial dynamics are less satisfactory than national aggregates (Section 3.3 and Appendix C). The headline results are averages over ten matched simulations; formal Monte Carlo uncertainty bands for all reported statistics are left to future work, and the historical benchmark is currently a univariate AR(1); multivariate VAR benchmarks are in progress.

Taken together, these limitations imply that MacroABM-CA is best used, at its current stage, to compare mechanisms, regional patterns and relative magnitudes across policy designs under clearly stated assumptions, rather than to produce point forecasts of programme costs or outcomes.

8 Conclusion

MacroABM-CA is, to our knowledge, the first open-source regional macroeconomic agent-based model of Canada. It is released with a newly constructed 2022 interprovincial input-output system spanning all thirteen provinces and territories in a common fifty-sector classification, which can serve as a reusable foundation for Canadian regional analysis independent of the model itself. The model regionalizes the MacroABM framework of Wiese et al. (2024) so that a national policy is transmitted through 650 province-sector firms that differ in scale, technology, sourcing and initial conditions, and it separates planned from realized quantities at every stage. Its historical performance, documented in Section 3 and Appendix C, is competitive with a simple statistical benchmark at the national level.

The investment tax credit examined here demonstrates what this framework adds. It shows how a single, explicitly stated firm-level assumption is reshaped by market realization, imported inputs, regional production structure and adaptive expectations: a substantial share of the stimulus leaks abroad through imported machinery, the response concentrates in the largest provinces while varying in intensity across smaller ones, and a temporary credit continues to affect activity after it ends. These are the questions that matter in a federation, where policy debate turns on where effects arise and to whom as much as on their aggregate size.

Validation in this framework is an ongoing process rather than a completed step (Section 3 and Appendices C and F). Three lines of development now under way will strengthen the model on precisely the grounds identified in Section 7: deeper firm heterogeneity disciplined by Canadian firm-level microdata, an endogenously adjusting production network, and fuller fiscal and monetary policy engines. Each will sharpen the empirical foundation of the regional transmission that MacroABM-CA is built to trace, while preserving its open and reproducible design.

Appendix

The appendices document the implementation detail, data construction, validation evidence and diagnostics required to interpret and reproduce the results in the main text. Appendix A records the implementation and parameter provenance; Appendix B describes the 2022 input-output data and the initial economic structure; Appendix C reports historical validation detail; Appendix D sets out the policy mechanism and its decomposition; Appendix E reports the regional, sectoral and network-exposure heterogeneity analysis; and Appendix F collects the robustness and parameter-sensitivity results. Replication code, configurations and data-construction procedures are available at <https://github.com/uvic-sesit/macroabm-ca>.

A Model implementation and parameter provenance

This appendix records the within-quarter event sequence and the parameters of the reported specification, so that the equations of Sections 2 and 4 can be read without the full software architecture.

A.1 Quarterly sequence

Each quarter proceeds through the four stages of Figure 1. The full ordered event list is:

Stage 1. Planning. Inherit balance sheets, stocks, prices, employment and realized-demand history; update macro and firm demand expectations; set target production; derive labour demand, intermediate orders and desired capital purchases. When the investment tax credit is active, its multiplier enters here.

Stage 2. Labour matching and production. Post vacancies and separations; match workers across industries subject to reservation wages and availability; produce the minimum of target, labour-, intermediate- and capital-feasible output; set prices.

Stage 3. Market realization. Firms, households and governments place orders; eligible agents apply for credit; credit is allocated subject to bank capital and borrower conditions; goods markets clear against production, inventories and foreign supply, rationing when supply is short; imports and interprovincial deliveries are realized.

Stage 4. Accounting and state update. Record sales, incomes, taxes, transfers and any tax-credit refunds; update inventories, capital stocks, financial positions and demand histories; carry states into the next quarter.

Capital purchased in a quarter is installed after that quarter’s production and becomes available subsequently; it cannot raise current output. Matched treatment and control simulations use common random-number streams for price setting and labour matching, reducing sampling noise in treatment-control differences.

A.2 Parameters of the reported specification

Table A1 lists the parameters that define the reported specification and classifies each by provenance: (1) data-implied from Canadian sources; (2) inherited from a formal calibration in the parent model; (3) inherited as a fixed or literature-based model assumption; (4) a Canadian validation or configuration choice; and (5) a policy-experiment assumption. The investment-response elasticity is a policy-experiment assumption and is externally specified rather than calibrated within the model.

Table A1. Parameters of the reported specification and their provenance.

Parameter	Symbol	Value	Role	Provenance
Regions and sectors		13 x 50 = 650 cells	province-sector representative producers	(1)
Firms per cell		1	representative producer; heterogeneity is structural	(4)
ITC statutory rate	tau	0.30	share of eligible expenditure refunded	(5)
Investment-response elasticity	eta_I	0.50	converts the eligible-share-weighted user-cost reduction into desired capital; externally specified	(5)
Eligible sectors	E	C27, C28	machinery-and-equipment proxy for eligible assets	(5)
Policy window		2026Q1 to 2030Q4 (20 quarters)	active-credit period	(5)
Credit design		project-wide multiplier, applied before the stock comparison	complementarity of eligible and other project capital	(5)
Refund basis		fully refundable credit equal to 30 percent of realized eligible purchases	fiscal booking of the credit	(5)
Monetary environment (main)		constant policy rate	monetary-policy setting	(4)
Monetary environment (robustness)		smoothed, inflation-only Taylor-type rule	alternative monetary-policy setting	(4)
Price demand-response coefficient	gamma_d	0.05	demand-pull price adjustment	(4)
Price cost-push coefficient	gamma_c	0 (excluded)	direct cost-push adjustment switched off	(4)
Household consumption rule		consumption out of disposable income	links employment and transfers to demand	(4)
Household investment		exogenous path	modelling choice	(4)
Government consumption		exogenous target, realized in the goods market	modelling choice	(4)

Parameter	Symbol	Value	Role	Provenance
Transfer scope		OECD SOCX public cash benefits	excludes in-kind services	(1)
Demand-smoothing speed	α	0.40	historically assessed adaptive-expectations adjustment speed; not formally estimated	(4)
Labour-force timing		annual level aligned to calendar year; membership updated before matching	projection and matching sequence	(4)
Labour and intermediate planning		follow target production without inherited-capital pre-clamping	preserves desired-versus-feasible distinction	(4)
Demand-growth weights	ϕ_g, ϕ_i	inherited (Wiese et al. 2024)	aggregate and firm-specific growth signals in expected demand	(3)
Unmet-demand weight		inherited (Wiese et al. 2024)	share of unmet orders re-entering recorded demand	(3)
Capital stock-adjustment rule	λ_K	inherited (Wiese et al. 2024)	partial adjustment toward the reference capital stock	(3)
Bank-rate pass-through		inherited (Wiese et al. 2024)	policy rate to lending and deposit rates	(2)
Technical coefficients	$\hat{a}^M, \hat{a}^K$	data-implied	fixed-proportion intermediate and capital requirements	(1)
Bilateral sourcing shares		data-implied	within-province, interprovincial and international sourcing	(1)

Notes: inherited behavioural parameters retain the parent-model configuration of Wiese et al. (2024) and, through it, Poledna et al. (2023); where a mechanism is unchanged the parent value is used and is not re-estimated for Canada. Parameters that the parent study calibrates formally are inherited at their calibrated values.

A.3 Selected functional detail

Reference capital stock and zero-output guards. The stock-adjustment term of Eq. (3) uses a rolling reference $K_{ij,t}^{\text{ref}} = K_{ij,t-1}(Q_{i,t}^*/Q_{i,t-1})$ with $\lambda_K = 0.1$. Two guards protect the recursion. If

previous output is zero, $Q_{i,t-1} = 0$, the growth ratio is set to one, so $K_{ij,t}^{\text{ref}} = K_{ij,t-1}$, the adjustment term vanishes and desired purchases reduce to replacement. If the inherited stock of a required capital type is itself zero, $K_{ij,t-1} = 0$, the rolling reference would lock the stock at zero, so for those types the model reverts to a base-year reference $K_{ij,t}^{\text{ref}} = (Q_{i,t}^*/Q_{i,0})K_{ij,0}$, which yields a positive rebuild. Both guards are diagnostic and do not otherwise alter behaviour. Target production is separately floored at a small positive value (Eq. 2), and the firm-specific growth term uses the same guard on previous supply.

Household consumption rule. The general rule is a three-branch maximum, allocated across sectors by household expenditure weights:

$$C_{h,t}^d = \frac{1}{1 + \tau_{vat}} \max \left\{ \phi_{\min}(1 - s_h)B_{h,t}, (1 - s_h)Y_{h,t}^d, \phi_c(1 + \tau_{vat})^{\frac{1}{w}} \sum_{k=1}^w C_{h,t-k} \right\},$$

where $B_{h,t}$ is benefit income, s_h the saving rate, $\phi_{\min}$ the minimum-consumption fraction, ϕ_c the smoothing weight and w the smoothing window. The reported specification sets $\phi_{\min} = 1$, $\phi_c = 0$ and $w = 12$, so the smoothing branch is inactive and the rule reduces to the two-branch form in Eq. (5). Expected transfers and expected financial income are the inflation-scaled counterparts of their realized values, and rental income enters net of income tax. Consumption is formed from expected income at the planning stage, before goods-market realization.

Taylor-type monetary rule. The alternative monetary environment sets the common rate on an annual-effective basis and converts it to the model’s quarterly-effective convention:

$$i_t^{\text{ann}} = \min \left\{ \bar{i}, \max \left\{ 0, \rho i_{t-1}^{\text{ann}} + (1 - \rho)[r^* + \pi^* + \phi_{\pi}(\pi_t^{\text{ann}} - \pi^*)] \right\} \right\},$$

with interest-rate smoothing $\rho = 0.85$, inflation response $\phi_{\pi} = 1.5$, neutral real rate $r^* = 0.005$, inflation target $\pi^* = 0.02$, a lower bound of zero and an upper bound $\bar{i} = 0.08$. The inflation term π_t^{ann} is the model’s current realized inflation expressed at an annual-effective rate; there is no output-gap term. The rule uses the previous policy rate, is initialized from the model’s initial policy rate, and its output is converted back to the quarterly-effective model rate.

B The 2022 Canadian economic structure underlying initialization

MacroABM-CA is initialized from a newly constructed 2022 interprovincial input-output (IO) system that reconciles Statistics Canada’s provincial supply-and-use and interprovincial trade data to the fifty-sector OECD Inter-Country Input-Output (ICIO) classification (Yamano et al. 2023), with the rest of the world as an external region. This appendix documents the provincial and sectoral structure that the initialization encodes. The exhibits are the observed 2022 economy, not model output. Values are in 2022 Canadian dollars at basic prices (the valuation in which the system is balanced); because the basic-price convention carries net taxes on products in a separate valuation bridge, these totals are slightly below the corresponding market-price national-accounts aggregates.

Aggregating the system over the thirteen provinces and territories reproduces the national totals to the level expected from the accounting scope. Gross value added (GVA) at basic prices for the thirteen provinces and territories is \$2,674.2B; adding the small extra-territorial enclave region that national accounting retains but the thirteen-region behavioural model excludes (\$0.8B) recovers the national control of \$2,674.95B, so the two figures differ only by that geographic scope and

are otherwise identical. Gross output for the thirteen provinces and territories is \$5,029.4B. National household final consumption is \$1,330.0B, government final consumption \$579.7B, gross fixed capital formation \$624.4B, international exports \$965.8B and international import use \$902.5B. Interprovincial balances net to zero across regions by construction.

B.1 Provincial structure

Table A2 reports each province's value added, expenditure-side final demand and trade balances. Value added is measured on the industry (production) side; household, government and investment demand are measured on the expenditure side as final demand attributed to the consuming region from all sources (own province, other provinces and abroad). Because final demand is attributed by consuming region and by category, the expenditure-side decomposition into household consumption, government consumption and investment is well defined at the provincial level.

Table A2. Provincial 2022 economic accounts (2022 CAD billions, basic prices).

Province/territory	GVA	Hhld. cons. (C)	Govt. cons. (G)	Invest. (I)	Exports (int'l)	Imports (int'l use)	Net int'l trade	Net inter- prov.
Newfoundland & Labrador	39.1	17.0	9.6	8.2	18.4	9.9	8.5	-4.6
Prince Edward Island	8.6	5.2	2.8	2.0	2.3	2.2	0.1	-1.8
Nova Scotia	51.0	33.4	19.4	12.0	10.4	17.1	-6.7	-8.3
New Brunswick	40.4	25.1	13.9	9.0	20.3	26.1	-5.8	-2.7
Quebec	512.8	267.0	134.9	115.3	158.2	182.0	-23.7	2.0
Ontario	987.8	520.0	209.7	228.9	357.6	400.7	-43.1	39.0
Manitoba	81.9	44.4	24.7	16.8	21.9	24.5	-2.7	-3.8
Saskatchewan	110.8	39.8	19.5	19.7	54.4	23.8	30.6	-3.5
Alberta	456.4	165.2	67.6	104.4	216.4	113.1	103.4	8.7
British Columbia	372.2	208.6	69.9	104.3	101.7	99.4	2.3	-21.1
Yukon	3.7	1.6	2.1	1.4	0.3	0.9	-0.6	-1.0
Northwest Territories	5.1	1.8	3.2	1.0	2.3	1.5	0.8	-1.7
Nunavut	4.3	0.9	2.5	1.5	1.6	1.3	0.3	-1.1
Canada	2,674.2	1,330.0	579.7	624.4	965.8	902.5	63.3	~0

Notes: basic prices, 2022 CAD billions. GVA is producer-side; household consumption, government consumption and investment are consuming-region final demand from all sources. Net international trade is international exports less international import use; net interprovincial balance is sales to other provinces less purchases from them and sums to approximately zero across regions. Import use is the cell-consistent import-use total; it lies about five percent below the aggregate international-import control, a difference attributable to re-exports, inventory and valuation that the underlying data do not separate.

Figure A1 visualizes this structure. Panel (a) shows the concentration of activity: Ontario, Quebec, Alberta and British Columbia account for the large majority of national GVA and final demand. Panel (b) shows the very different external orientations of the provinces. Alberta (\$103.4B) and Saskatchewan (\$30.6B) are large net international exporters, reflecting energy and resource output; Ontario is a net international importer (-\$43.1B) but the largest net interprovincial exporter

(+\$39.0B), consistent with its role as a domestic manufacturing and distribution hub. These structural differences are the initial heterogeneity through which a nationally uniform ITC produces different regional outcomes.

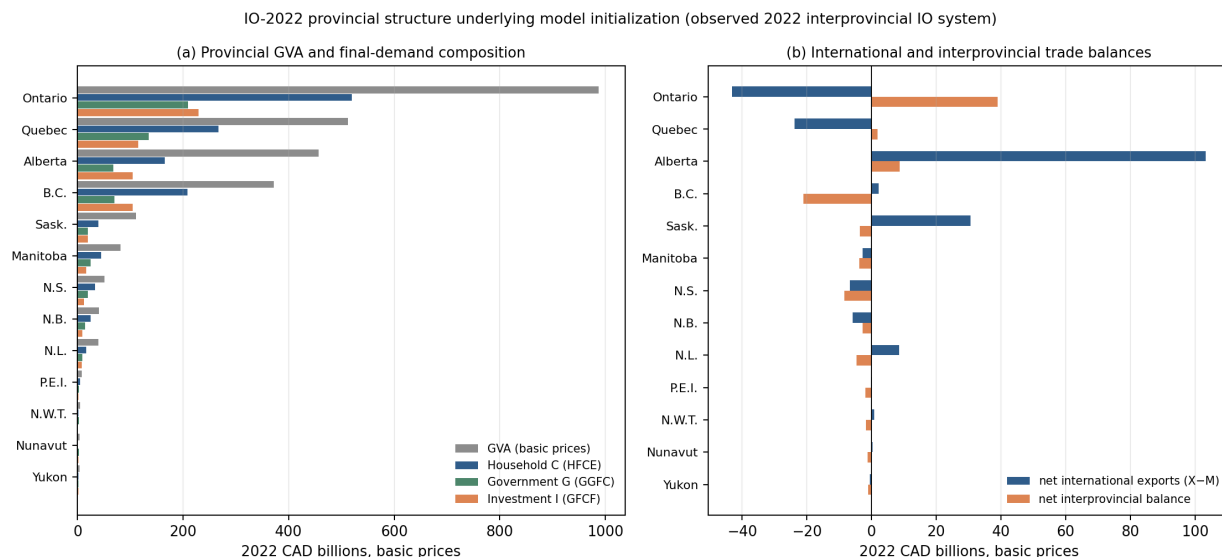


Figure A1. Observed 2022 provincial structure. Panel (a): provincial GVA (basic prices) and the household, government and investment components of final demand. Panel (b): net international trade (exports less international import use) and net interprovincial balance, in 2022 CAD billions.

B.2 Sectoral structure and the correct IO representation of final demand

At the sectoral level the expenditure-side decomposition used for provinces is not the economically correct object, and it is not imposed. Industries generate value added and supply commodities; they do not consume in the national-accounts sense. Household, government and investment demand are categories of final use satisfied by the output of many industries, not attributes of an industry, so a sectoral consumption, government or investment total would misrepresent the supply-and-use structure. The economically valid sectoral objects are: sectoral value added on the industry side; the sectoral supply of each final-demand category (which producing sectors supply household consumption, government demand and gross fixed capital formation); and sectoral trade exposure (exports and imported use by commodity).

Figure A2, panel (a), reports sectoral value added. The largest sectors by national GVA share are real estate (11.4 percent, including imputed rent), wholesale and retail trade (10.3 percent), public administration (7.9 percent), construction (7.6 percent), health and social work (7.2 percent), finance and insurance (6.6 percent) and oil and gas extraction (6.5 percent; the upstream mining sector producing crude oil and natural gas, distinct from electricity and gas distribution utilities, which form a separate sector), a profile characteristic of a services-and-resource economy.

The sectoral supply of final demand is category-specific: household consumption is supplied mainly by real estate (\$311.9B, largely imputed rent), trade (\$251.9B), finance (\$125.8B) and food and beverages (\$103.8B); government consumption is supplied predominantly by the health and public-administration industries; and gross fixed capital formation is supplied primarily by construction (\$391.4B), motor vehicles (\$33.3B), machinery n.e.c. (C28, \$30.5B) and real estate (\$32.5B). Do-

mestic industries supply \$638.9B of gross fixed capital formation against \$624.4B of investment demand, the difference reflecting net trade in capital goods and valuation.

Figure A2, panel (b), reports sectoral trade exposure and is directly relevant to the policy experiment. Oil and gas extraction is by far the largest net exporter (+\$141.0B), followed by wholesale and retail trade, non-ferrous metals, land transport and wood products. The largest net importers are computer and electronic products (-\$42.6B), motor vehicles (-\$39.1B), machinery n.e.c. (C28, -\$30.4B), chemicals and textiles. Both ITC-eligible sectors are net importers: electrical equipment (C27) has net international trade of -\$25.0B and machinery n.e.c. (C28) -\$30.4B, with imported use of \$33.9B and \$64.6B respectively. This structural feature is the initialization-level counterpart to the foreign-sourcing result of Section 5.6: because the eligible machinery sectors are heavily import-exposed in the 2022 data, an incentive that raises demand for their output draws in a substantial volume of imports.

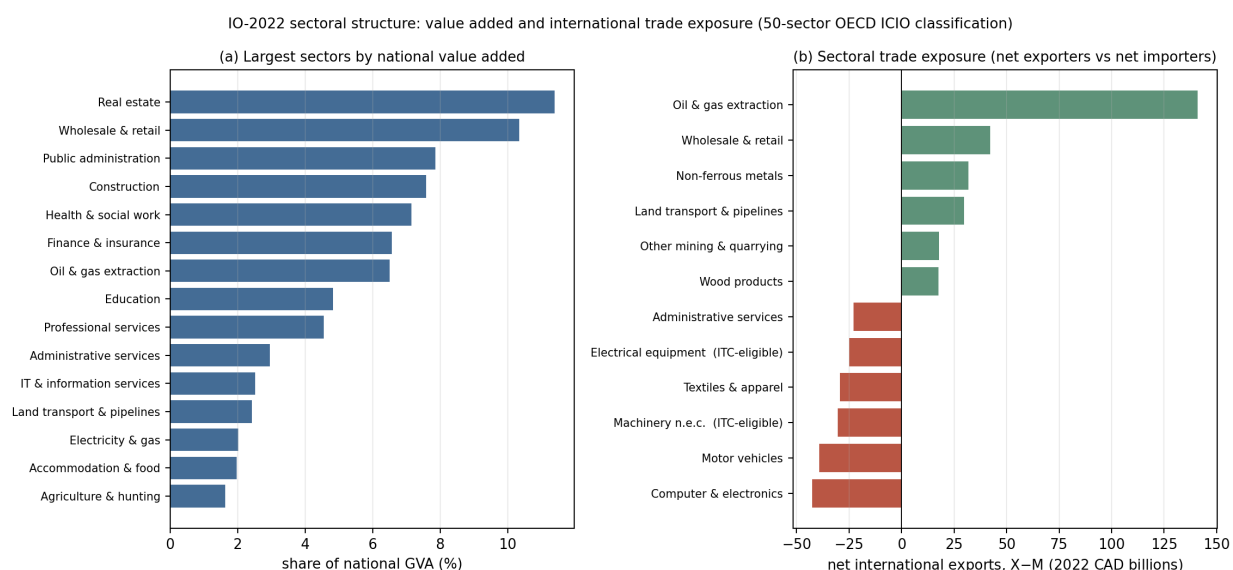


Figure A2. Observed 2022 sectoral structure (fifty-sector OECD ICIO classification). Panel (a): largest sectors by share of national GVA. Panel (b): sectoral net international trade (exports less imported use), showing the largest net exporters and net importers; the two ITC-eligible sectors (C27 electrical equipment, C28 machinery n.e.c.) are labelled and are both net importers.

B.3 Mapping the 2022 economy into the model structure, and IO-data limitations

Table A3 summarizes how the observed 2022 economy maps into the model representation used in the experiments.

Table A3. The 2022 IO economy mapped into the MacroABM-CA regional-sector structure.

Model object	2022 source	Resolution	Notes
Regions	Statistics Canada provincial economic accounts and interprovincial trade	13 provinces and territories, plus an external rest-of-world region	NL, PE, NS, NB, QC, ON, MB, SK, AB, BC, YT, NT, NU
Sectors	OECD ICIO classification	50 industries	harmonized from the native Statistics Canada industry detail
Representative firm cells	region by sector	650 producers	one representative firm per province-sector
Intermediate flows	observed bilateral basic-price commodity trade	producer cell by user cell	interprovincial trade attributed at origin sector and destination sector
Final demand	household, non-profit, government, fixed capital formation, inventories and direct purchases abroad	6 categories by consuming region and source	international source encoded as rest-of-world origin
Value added	wages, employer contributions, mixed income, operating surplus and net production taxes	region by sector	basic prices
Valuation and units	basic prices, 2022 current CAD		net product taxes carried in a separate valuation bridge
ITC-eligible proxy	C27 (electrical equipment) and C28 (machinery n.e.c.)	2 of 50 sectors	broad machinery-and-equipment proxy; statutory eligibility not reproduced

The interprovincial trade structure is constructed from observed data combined with one explicit allocation assumption, which we state precisely because it bounds the resolution of the regional analysis. What is observed is the bilateral province-to-province trade in each commodity: every province’s basic-price accounts record its interprovincial exports to, and imports from, each partner province by commodity, and these bilateral commodity totals are mutually consistent (each province’s recorded exports to a partner equal that partner’s recorded imports) without any balancing or matrix-fitting procedure. What is not observed, and therefore allocated, is the split of a destination province’s imports of a commodity across the industries and final users that absorb them by province of origin. We observe how each destination’s total imports of a commodity are used across its industries, and we assign the imports from each origin province to those using industries in the same proportions, that is, under an origin-homogeneity assumption that the origin mix is common across the using industries of a given commodity. The observed margins pin down the bilateral totals; the origin-homogeneity assumption fills only the interior of the origin-by-commodity-by-using-industry array. International imports are allocated to Canadian using industries with a comparable use-structure assumption. This is a genuine limitation: the model does not rest on fully observed origin-by-commodity-by-industry bilateral flows, which Canadian sources do

not provide at this resolution, and refinements from establishment-level trade surveys would relax it rather than replace it.

C Historical validation detail

This appendix supplements the national validation of Section 3.3 with the provincial scorecard and an external time-series benchmark. The historical exercise is a conditional historical simulation: selected external paths are supplied during 2022 to 2024, so even a close fit is not a clean out-of-sample forecast.

C.1 Provincial scorecard

The provincial diagnosis compares observed and simulated outcomes for the ten provinces over 2022 to 2024. National performance is stronger than equal-weight provincial performance because large provinces dominate national aggregates and offsetting provincial errors partly cancel. Simulated provincial GVA growth is compressed toward the national path relative to the observed cross-province dispersion. Provincial unemployment errors are disproportionately influenced by small provinces, where population and labour-force scaling leave fewer agents and greater sensitivity to stochastic matching. Common behavioural parameters and fixed production and sourcing coefficients also limit province-specific adjustment.

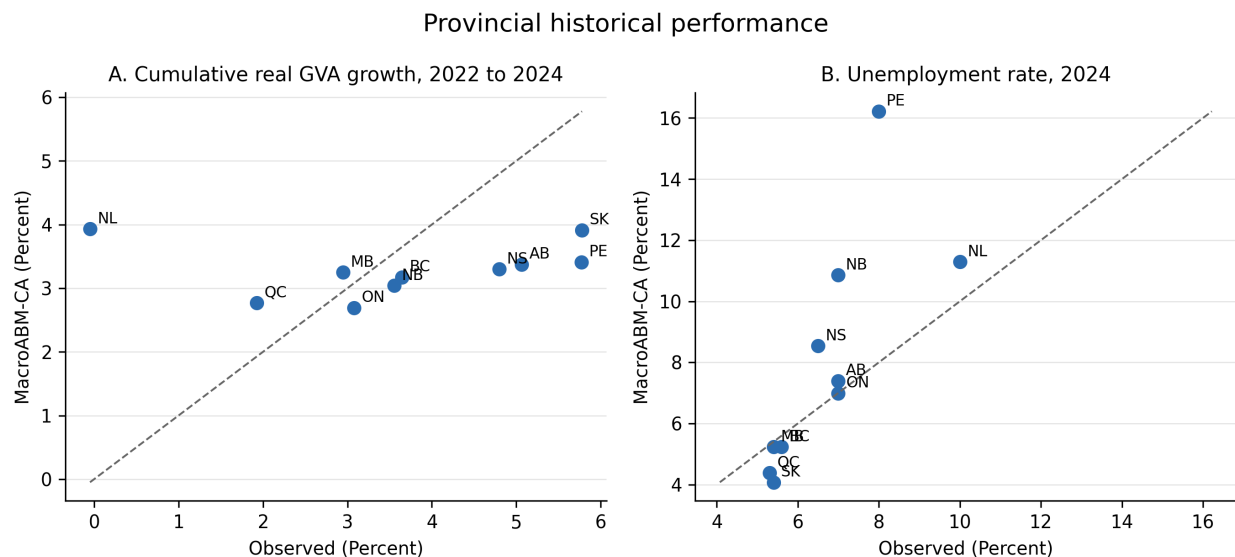


Figure A3. Observed and simulated provincial outcomes. Panel A compares cumulative real GVA growth from 2022 to 2024. Panel B compares 2024 unemployment rates. Points are simulation averages and the dashed 45-degree line denotes equality. The clustering of simulated GVA growth relative to the wider observed range illustrates compressed cross-province dispersion; the unemployment panel highlights the influence of small-province observations on equal-weight errors.

C.2 External time-series benchmark

Following the benchmarking practice of Poledna et al. (2023) and Wiese et al. (2024), national historical performance is compared with a deliberately parsimonious AR(1) benchmark. For each

variable, the benchmark is

$$x_t = c + \rho x_{t-1} + \varepsilon_t.$$

The real-output equation uses 100 times the quarterly log difference of Statistics Canada GDP at basic prices for all industries in chained 2017 dollars (table 36-10-0449-01, vector v1000000673). The series is seasonally adjusted at annual rates, and the estimation sample is 1997Q3-2021Q4. The unemployment equation uses the quarterly arithmetic mean of the seasonally adjusted monthly unemployment rate for Canada, both sexes, ages 15 and older (table 14-10-0287-01, vector v2062815). It is estimated in rate levels over 1976Q2-2021Q4. The corresponding estimates are $c = 0.6580$ and $\rho = -0.1329$ for real-output growth, and $c = 0.6450$ and $\rho = 0.9210$ for unemployment.

Both equations are forecast recursively from 2021Q4 through 2024Q4. Realized observations within that interval are not used to update either the coefficients or the forecast state. The source series are from the latest available official vintage, so this is a fixed-origin latest-vintage exercise rather than a real-time-vintage forecast evaluation. Real-output growth is scored over 2022Q2-2024Q4 because the saved MacroABM-CA output begins in 2022Q1 and therefore does not permit calculation of model growth for that first quarter. Unemployment is scored over all twelve quarters from 2022Q1 through 2024Q4.

The information sets are not identical. The AR(1) uses only its own pre-2022 history, whereas the MacroABM-CA historical exercise is conditional on selected observed provincial demand and labour-force paths during the evaluation period. Table A4 and Figure 2 therefore benchmark historical performance against a simple external statistical model; they do not constitute a strictly controlled forecasting horse race.

Table A4. National historical fit versus an AR(1) benchmark, 2022 to 2024 (root-mean-square error, RMSE, in the units of each series).

Variable	AR(1) RMSE	AR(1) MAE	MacroABM-CA RMSE	MacroABM-CA MAE	Relative assessment
Real GVA growth (percent per quarter)	0.3542	0.2941	0.4360	0.3635	MacroABM-CA errors 23.1%/23.6% larger
Unemployment rate (percentage points)	1.3392	1.2781	0.9219	0.7997	MacroABM-CA errors 31.2%/37.4% smaller

Notes: the first percentage in the final column refers to RMSE and the second to MAE. A model-versus-official CPI score is not reported because the model's internal price index is a realized-consumption-weighted index of domestic output prices at basic prices, whereas Statistics Canada's CPI is a purchaser-price fixed-basket consumer index that includes imports, shelter and indirect taxes.

D Policy mechanism and decomposition

D.1 Eligible share, multiplier and timing

For firm i , the lagged eligible share $s_{i,t-1}$ is the value share of the eligible sectors C27 and C28 in the preceding planned capital bundle. During the policy window the credit scales the requirement term of Eq. (3) before the stock comparison:

$$I_{ij,t}^d = \max \left\{ 0, m_{i,t} d_{ij} \tilde{Q}_{i,t}^K + \lambda_K (K_{ij,t}^{\text{ref}} - K_{ij,t-1}) \right\}, \quad m_{i,t} = (1 - \tau s_{i,t-1})^{-\eta_I},$$

with $m_{i,t} = 1$ outside the window. Three points make the mechanism precise. First, $\tilde{Q}_{i,t}^K = \min\{Q_{i,t}^*, \mathcal{C}_{i,t-1}^K\}$ is unchanged by the credit: the multiplier does not replace $\tilde{Q}^K$ with Q^* . Second, $m_{i,t}$ scales only the requirement term $d_{ij} \tilde{Q}^K$, which includes the replacement of depreciated capital; the stock-gap term $\lambda_K (K^{\text{ref}} - K_{t-1})$ is not multiplied by m . Third, the multiplier enters before the desired requirement is compared with the inherited stock, so it raises desired gross purchases while preserving the baseline production, requirement and stock-adjustment structure. When a firm has no eligible capital in its planned bundle the eligible share is zero, the multiplier is one, and the firm receives no user-cost reduction and no refund. The refund is computed on realized eligible purchases, not on desired orders: purchases must clear the goods market, so rationing and sourcing determine both the installed capital and the refund. Purchased capital is installed after the current quarter's production and enters feasible production only in subsequent quarters. Refunds are booked once, in the quarter of realized eligible purchase, and are zero after 2030.

D.2 Same-state investment diagnostic

The same-state diagnostic isolates the contemporaneous user-cost effect by evaluating the model with and without the ITC multiplier at an identical inherited state. It separates the increment produced by the current multiplier from differences in capital stocks, prices, eligible-capital shares, demand histories and other states inherited from earlier periods.

D.3 Channel A and Channel B decomposition

The channel decomposition uses an own-demand replay intervention on the treatment-control comparison. Channel A, the fixed-expectations channel, is the treatment-control response when firms' own-demand histories are replayed from the control simulation, so that the policy cannot feed back through firms' own recorded demand, while the ITC, current transactions, household behaviour, prices and all other feedbacks remain active. Channel A is therefore not a pure direct, physical or capacity effect: it retains every mechanism except the policy-induced updating of firms' own-demand histories. Channel B, the adaptive-demand channel, is the full treatment-control response less Channel A, and measures the contribution associated with allowing policy-affected sales and own-demand histories to enter subsequent expectations, including their interaction with the other active mechanisms. The decomposition is intervention-based and is not a uniquely additive structural partition.

Table A5. Channel A and Channel B contributions (simulation averages; cumulative; all quantities in constant prices, following Table 3).

Cumulative response (constant-price \$B)	Channel A	Channel B	Channel B share (percent)
Investment, 2026-2030	22.11	14.71	40.0
Production, 2026-2030	46.30	57.60	55.4
GVA, 2026-2030	21.67	27.66	56.1
Investment, 2031-2035	1.33	44.22	97.1
Production, 2031-2035	48.41	224.08	82.2
GVA, 2031-2035	22.80	110.47	82.9

Notes: shares are ratios of average Channel B responses to average full responses, rather than averages of simulation-level shares.

Adaptive updating accounts for slightly more than half of the active-window production and GVA responses and about four-fifths of the post-policy responses. Because the adaptive contribution develops in an economy whose stocks and transactions have already changed, it does not imply that capital accumulation is unimportant.

E Regional, sectoral and IO-exposure heterogeneity

Section 5.4 examines cross-cell transmission heterogeneity using the province-sector cell as the unit of observation. The principal response measure divides the cumulative 2026-2030 real-production effect by twenty quarters of average pre-policy production. The direct investment impulse is normalized analogously by pre-policy investment. Cells with denominators below 10^{-8} of the corresponding national maximum are excluded from ratio calculations, and the first and ninety-ninth percentiles are used only to limit the displayed histogram. The underlying observations remain unwinsorized.

The interpretation distinguishes propagation over the fixed IO structure from endogenous network formation. Because the experiments use one representative firm per province-sector cell, network position is largely a function of sector and province. A first-order IO analysis could augment the own-exposure measure with a sales-share-weighted customer exposure and a purchase-share-weighted supplier exposure. Such exposure measures would be descriptive rather than causal, and constructing them over the fixed input-output system of the reported specification is left to future work.

E.1 Provincial policy scorecard

Table A6 reports the full provincial scorecard underlying the regional discussion of Section 5.4.

Table A6. Provincial 2026-2030 treatment-control effects of the investment tax credit (simulation averages).

Province	Refunds (\$B)	Investment (\$B)	Production (\$B)	GVA (\$B)	Employ- ment (persons)	Unemp. rate (pp)	Imports (\$B)	GVA per refund	GVA / pre- policy scale (%)
NL	0.71	0.53	0.93	0.56	1,300	-0.364	0.22	0.80	0.28
PE	0.21	0.18	0.51	0.27	100	-0.125	0.08	1.24	0.62
NS	0.83	0.67	1.90	0.97	1,300	-0.306	0.30	1.17	0.38
NB	1.16	0.81	1.89	1.10	2,900	-0.879	0.23	0.94	0.53
QC	8.88	6.42	20.61	9.70	22,800	-0.518	2.92	1.09	0.37
ON	16.66	12.04	37.13	16.53	51,700	-0.501	6.64	0.99	0.33
MB	1.39	0.88	2.30	1.20	4,300	-0.494	0.37	0.86	0.29
SK	4.41	2.70	6.15	3.44	13,000	-1.335	1.18	0.78	0.60
AB	11.57	8.96	24.03	11.07	39,400	-0.894	2.85	0.96	0.47
BC	4.42	3.39	7.67	4.10	7,500	-0.229	0.94	0.93	0.22
YT	0.07	0.08	0.19	0.11	201	-0.514	0.03	1.48	0.56
NT	0.12	0.08	0.15	0.09	174	-0.383	0.05	0.77	0.34
NU	0.21	0.21	0.45	0.20	379	-0.995	0.08	0.97	0.90
Canada	50.14	36.82	103.91	49.33	145,054	-0.259	15.86	0.98	0.35

Notes: treatment minus control, averages across matched simulations, cumulative over 2026-2030 for flows and end-2030 differences for employment. Production and GVA are valued at fixed initialization prices; refunds, investment and imports are deflated to constant prices by each region's GVA deflator. Because that deflation is region-specific, provincial refund, investment and import flows need not sum exactly to the national totals, which are deflated by the national GVA deflator and match Table 3. Gross refunds are treatment levels, not net fiscal costs. The pre-policy normalization divides the cumulative GVA effect by twenty times average quarterly control GVA in 2024-2025. The national unemployment effect is computed from national employed and labour-force totals and is therefore not the sum or simple average of the provincial percentage-point effects. Response-per-refund ratios for the smallest provinces and the territories rest on small absolute bases and are correspondingly sensitive.

E.2 Province-sector structural exposure

At the province-sector level, the normalized production response shows a modest positive unconditional association with the pre-policy international import share (Spearman rank correlation 0.26) and essentially no unconditional association with the pre-policy interprovincial-sourcing share (Spearman 0.04). Figure A4 plots these relationships. Both are descriptive correlations over fixed sourcing coefficients; they do not identify import leakage, domestic multipliers or causal interprovincial spillovers, and they are not used to make bilateral interprovincial-flow claims.

Structural exposure and province-sector policy transmission

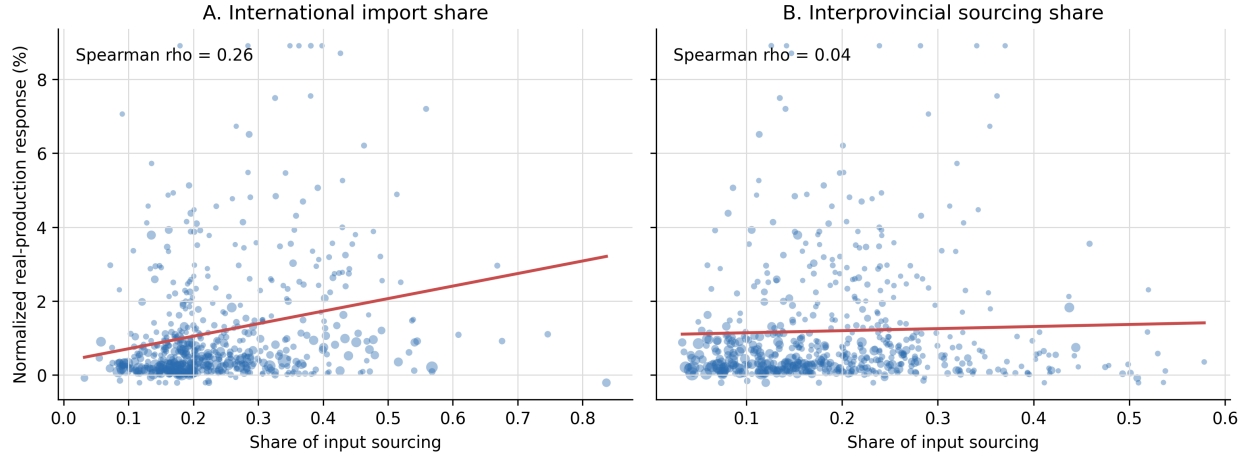


Figure A4. Province-sector normalized 2026-2030 production response against pre-policy international import share (left) and pre-policy interprovincial-sourcing share (right), across the 650 cells with stable pre-policy denominators. The association with import share is modest and positive; the association with interprovincial-sourcing share is negligible. These are descriptive correlations, not causal spillover estimates.

F Robustness and parameter sensitivity

F.1 Simulation uncertainty

The principal results in Table 3 are averages across matched simulations. The active-window investment, production and GVA responses are positive, while longer-horizon outcomes are more dispersed. Figures 3 and 7 show this dispersion as the tenth-to-ninetieth-percentile range around the simulation average. Post-policy dispersion is substantially wider than active-window dispersion. The long-horizon current-price capital-stock difference is especially sensitive: because capital is valued at contemporaneous prices, it reflects treatment-control differences in the price level rather than physical capital alone, and it is therefore not reported in Table 3. Post-policy magnitudes should be read as indicative of persistence rather than as precise estimates.

F.2 Monetary environment

The monetary robustness exercise replaces the constant policy rate with the stylized Taylor-type rule described in Appendix A. All other model mechanisms and policy parameters are held fixed. Figure A5 compares the 2026-2030 Channel A and Channel B responses for investment, production and GVA. Channel A production and GVA are nearly invariant across the two monetary environments. The policy and lending-rate treatment-control responses are small, and differences in the aggregate responses are concentrated mainly in Channel B. These differences are not interpreted as a systematic estimate of monetary attenuation or amplification; the exercise is a stylized robustness check rather than a calibrated Bank of Canada experiment. The result instead demonstrates that the ITC mechanism remains operative when the model's endogenous monetary-policy capability is activated and that monetary feedback interacts primarily with the adaptive-demand channel.

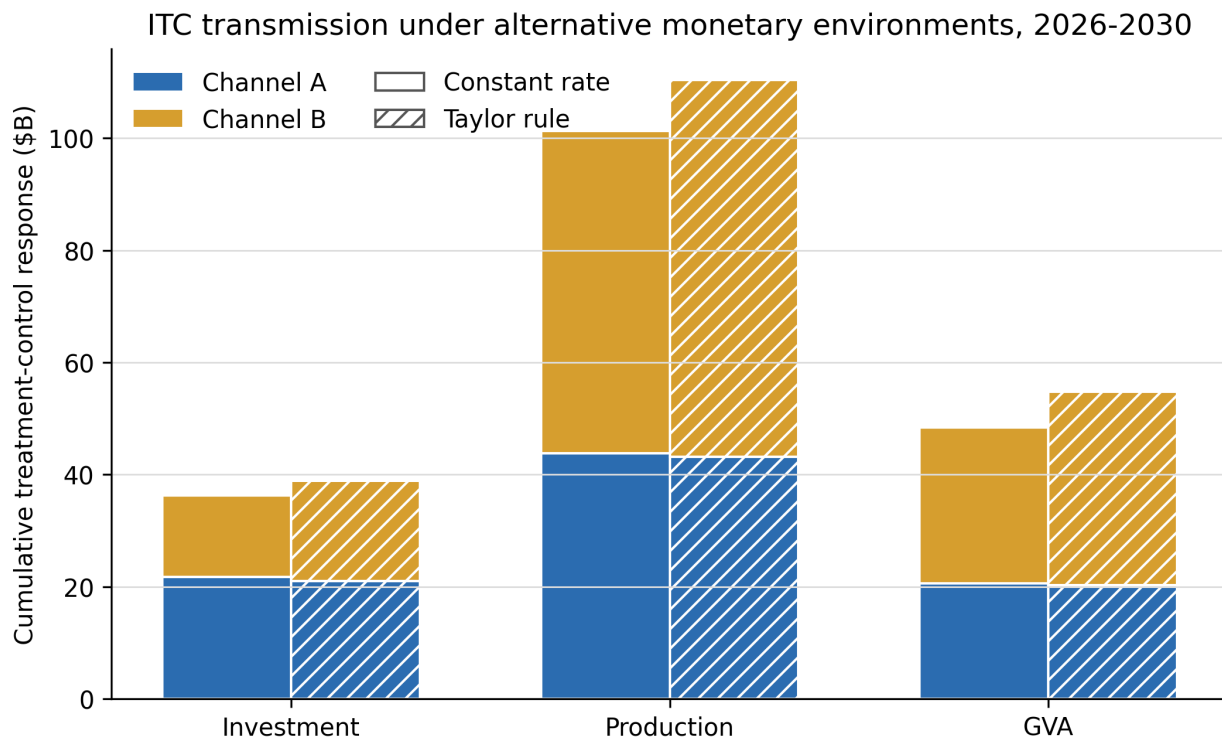


Figure A5. Investment-tax-credit transmission under constant-rate and Taylor-type monetary environments, 2026-2030. Bars report cumulative treatment-control responses and decompose each total into Channel A and Channel B. Solid bars denote the constant-rate environment and hatched bars the Taylor environment. Channel A production and GVA are almost identical across regimes, while differences in total responses arise mainly through Channel B. The figure is a robustness demonstration of endogenous monetary feedback, not a calibrated Bank of Canada forecast or an estimate of systematic monetary amplification.

F.3 Investment-response elasticity

The investment-response elasticity η_I is externally specified rather than estimated within the model, and the main results set it to 0.50. The sensitivity exercise holds all other features fixed and varies the elasticity over 0.20, 0.35, 0.50, 0.75 and 1.00, so it isolates the scaling of the response with η_I . Figure A6 shows approximately linear scaling of investment, production and GVA, with mild concavity at the upper end. Dividing each response by η_I yields values that decline only modestly at higher elasticities: investment falls from \$66.10 billion per unit of η_I at 0.20 to \$63.04 billion at 1.00, production from \$175.55 billion to \$167.29 billion and GVA from \$84.61 billion to \$81.85 billion. Channel B remains important throughout, accounting for 49.8 to 59.2 percent of GVA.

Investment-tax-credit user-cost elasticity sensitivity

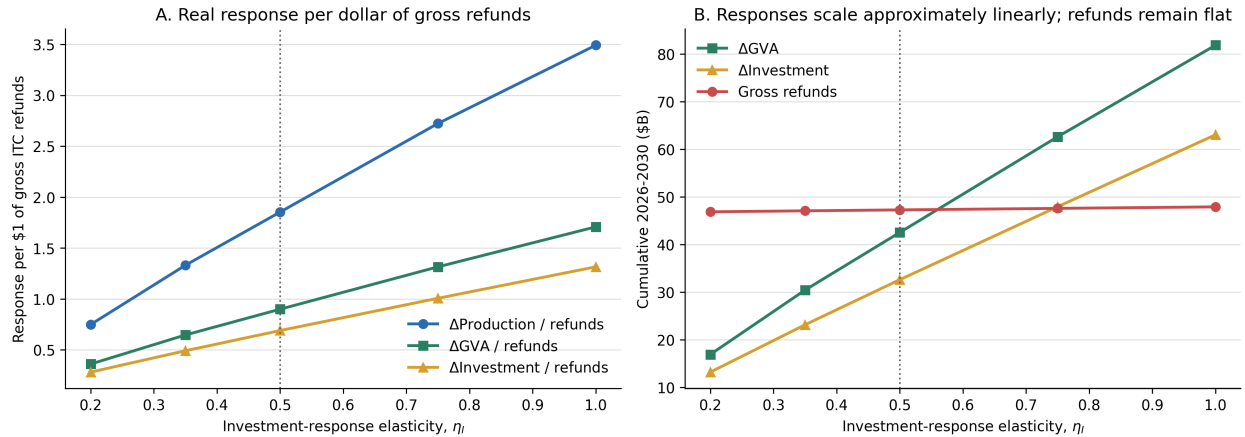


Figure A6. Sensitivity to the investment-response elasticity, 2026-2030. Panel A reports the cumulative treatment-control investment, production and GVA responses per dollar of gross ITC refunds. Panel B reports the cumulative investment and GVA responses together with gross refunds. Real responses scale approximately linearly with η_I , with mild concavity at higher values, while gross refunds remain nearly flat because they include inframarginal eligible purchases. Gross refunds are distinct from net fiscal cost, and response-per-refund measures are not fiscal multipliers or benefit-cost estimates.

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